



UNIVERSITÀ DEGLI STUDI
DI MILANO

Corso di
Visione Artificiale

Laurea Magistrale in Informatica (F94)

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Where are we?

First part:

Image formation and Early vision

- Image formation
 - Geometric Camera Models
 - Color spaces
- Image Processing
 - Punctual and spatial processing
 - Feature Extraction
- Reconstruction
 - Camera calibration
 - **Stereo Vision**
 - Structure from Motion and RGB-d Cameras
- Optical flow and Tracking

Second part:

Machine learning for CV

- Linear Neural Network
- Multi Layer Perceptron
- Convolutional Neural Networks
- Recurrent Neural Networks
- Transformers
- Variational Auto-Encoders
- Generative Adversarial Networks
- Graph Neural Networks
- Self-supervised learning
- Vision Language Models

STEREO VISION

Chapter 7 – Forsyth Ponce

Slide credits: L. Cinque, F. Pedersini, S. Nayar

What is stereo vision? Generic problem formulation

- Given several **images of the same object or scene**, compute a **representation of its 3D shape**
- **we need “Images of the same object or scene”**
 - An arbitrary number of images (from two to thousands)
 - Arbitrary camera positions (isolated cameras or video sequence)
 - Cameras can be calibrated or uncalibrated
- **“Representation of 3D shape”**
 - Depth maps
 - Meshes
 - Point clouds
 - Patch clouds
 - Volumetric models
 - Layered models

What is stereo vision? Narrower formulation

- Given a calibrated binocular stereo pair, fuse it to produce a **depth image**.
- We implement the **backward projection**, from 2D to 3D

left
image



right
image

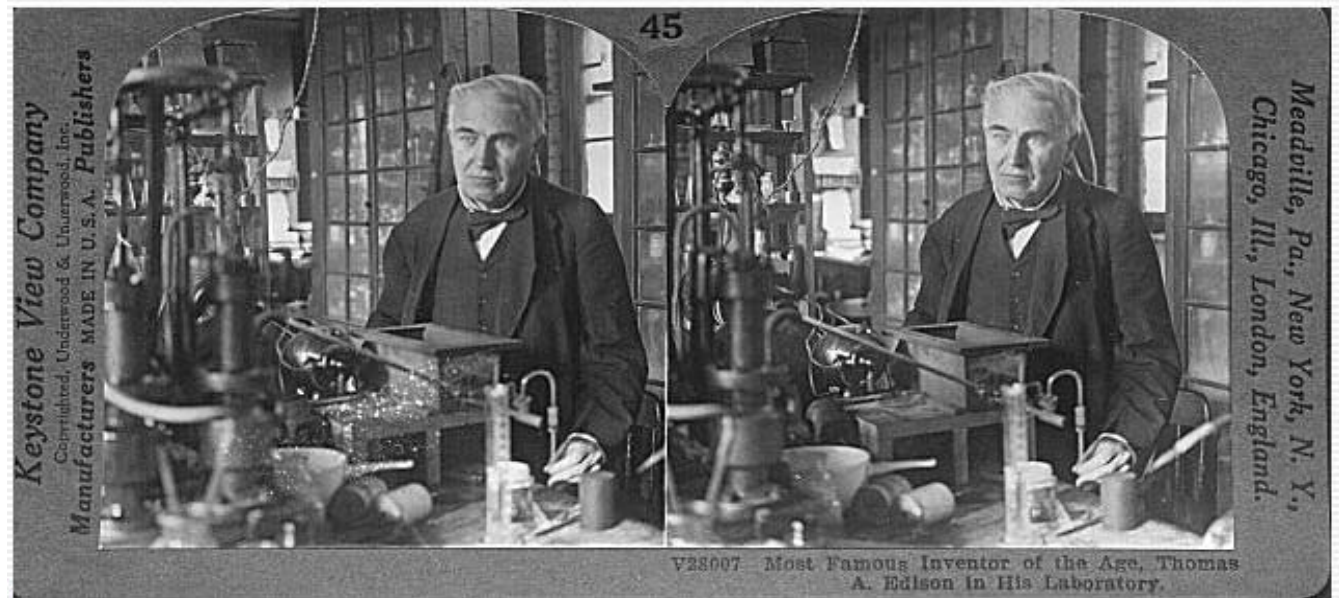


Depth
map

Human stereopsis

- Narrower formulation: given a calibrated binocular stereo pair, fuse it to produce a **depth image**

→ **Human can do it!**



Stereograms:

Invented by Sir Charles Wheatstone, 1838

Devices for stereopsis: 3D glasses

3D glasses work by filtering two different images for each eye, displayed on an external screen.

- **Anaglyph** (Red-Cyan) Separates images using colors.
- **Polarized** (Cinema 3D): Uses optical filters.
- **Active Shutter** (TV 3D): Uses LCD shutters synchronized with the display.

✗ Limitation: Creates a 3D effect on a 2D screen, but lacks immersion.



Devices for stereopsis: 3D headsets

3D headsets completely cover the field of vision and display different images for each eye.

- **VR Headsets:** Fully immersive digital environments (e.g., Oculus Quest, HTC Vive, PlayStation VR).
- **AR Headsets:** Overlay digital elements onto the real world (e.g., Microsoft HoloLens, Magic Leap).

✓ **Advantages over 3D glasses:**

- 360° immersive and interactive experience
- Motion tracking and real-world interaction
- Compatible with controllers and hand tracking

✗ **Disadvantages:**

- More expensive and requires high-performance hardware
- Can cause motion sickness in some users



View Interpolation



Left image

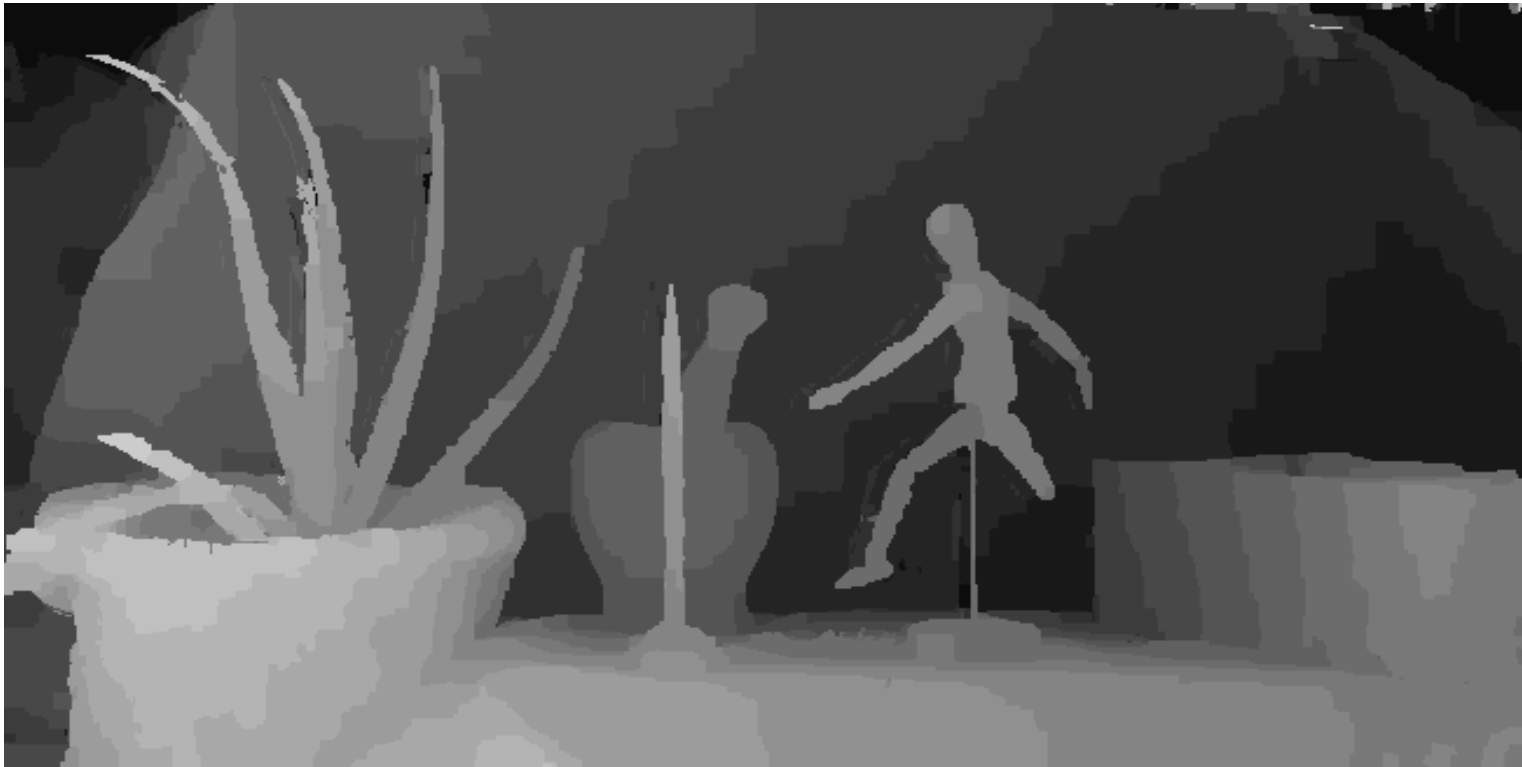
View Interpolation



Right image

View Interpolation

Disparity: the **difference in the image position** of the same 3D point observed from two different viewpoints



Light intensity values: greater disparity, closer objects

Disparity ($\rightarrow$ distance?)

View Interpolation

Disparity as a function of the object distance.



Interpolation between left and right views
using depth information

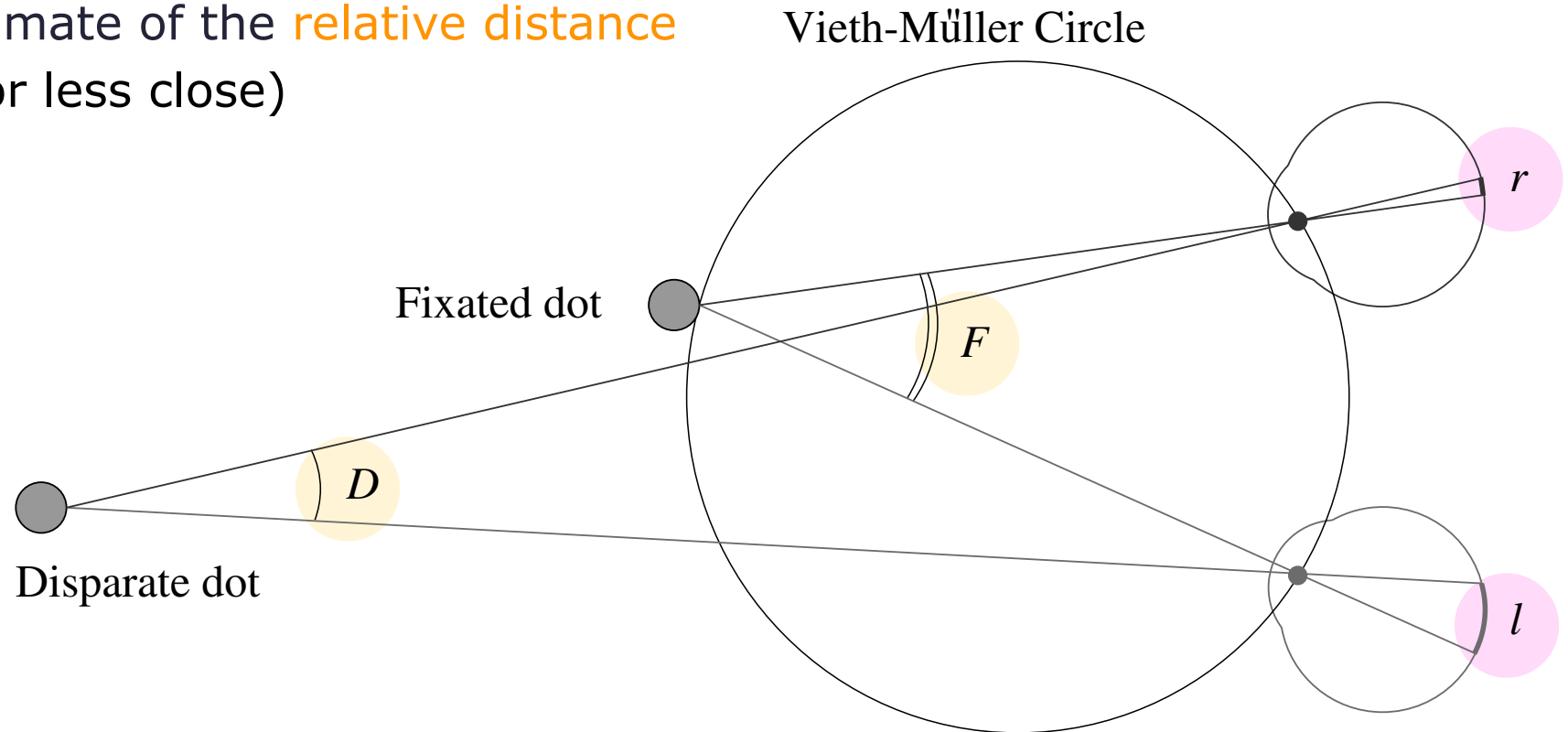
Human stereopsis

- Visual perception based on angular disparity.
Circle of **Vieth-Müller**: place of points to zero disparity → **same angular distance**

→ Not good estimate of the **absolute distance**

→ Good estimate of the **relative distance**

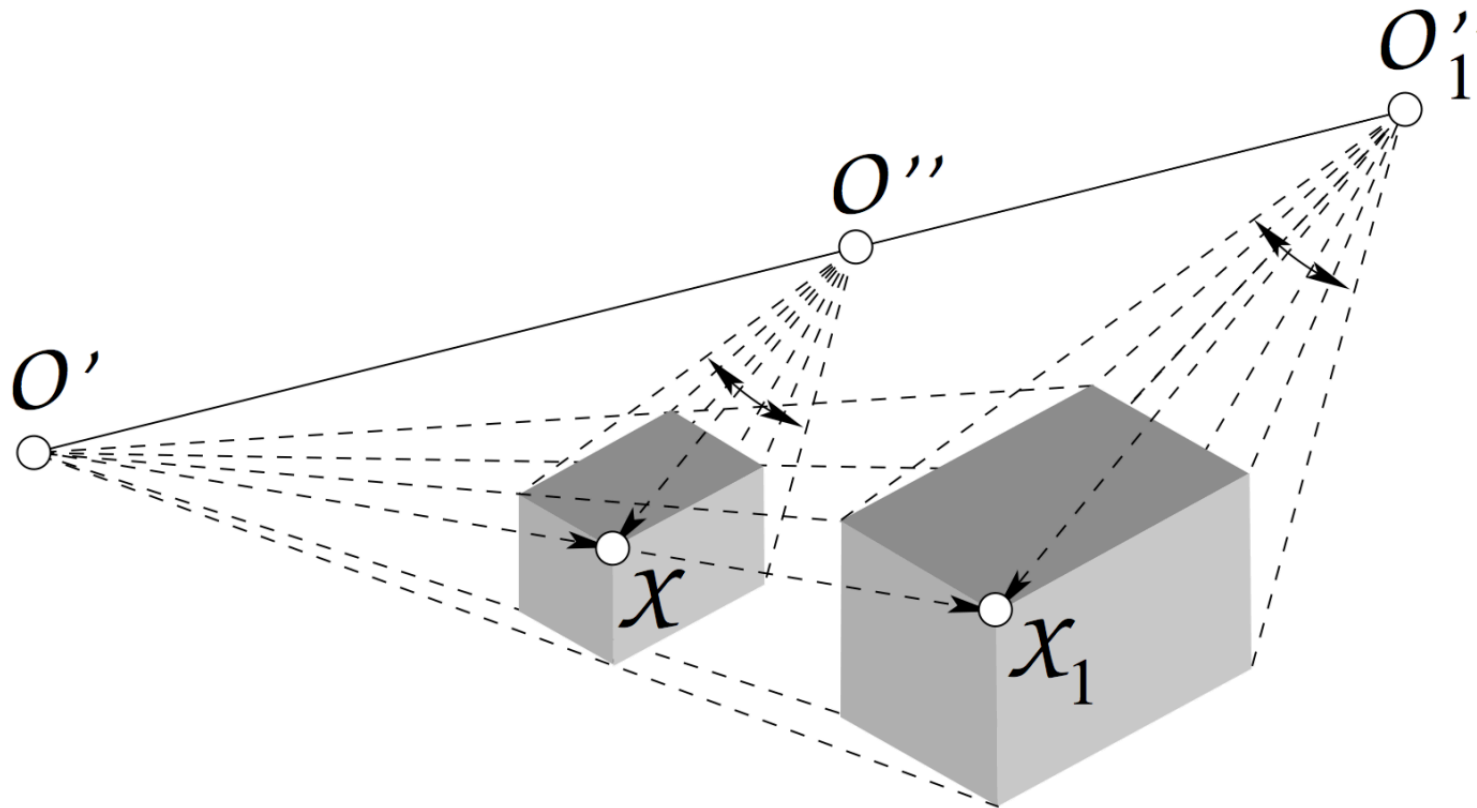
➤ (more or less close)



- **Eye movement** improve the visual perception of the scene compensating for the disparity

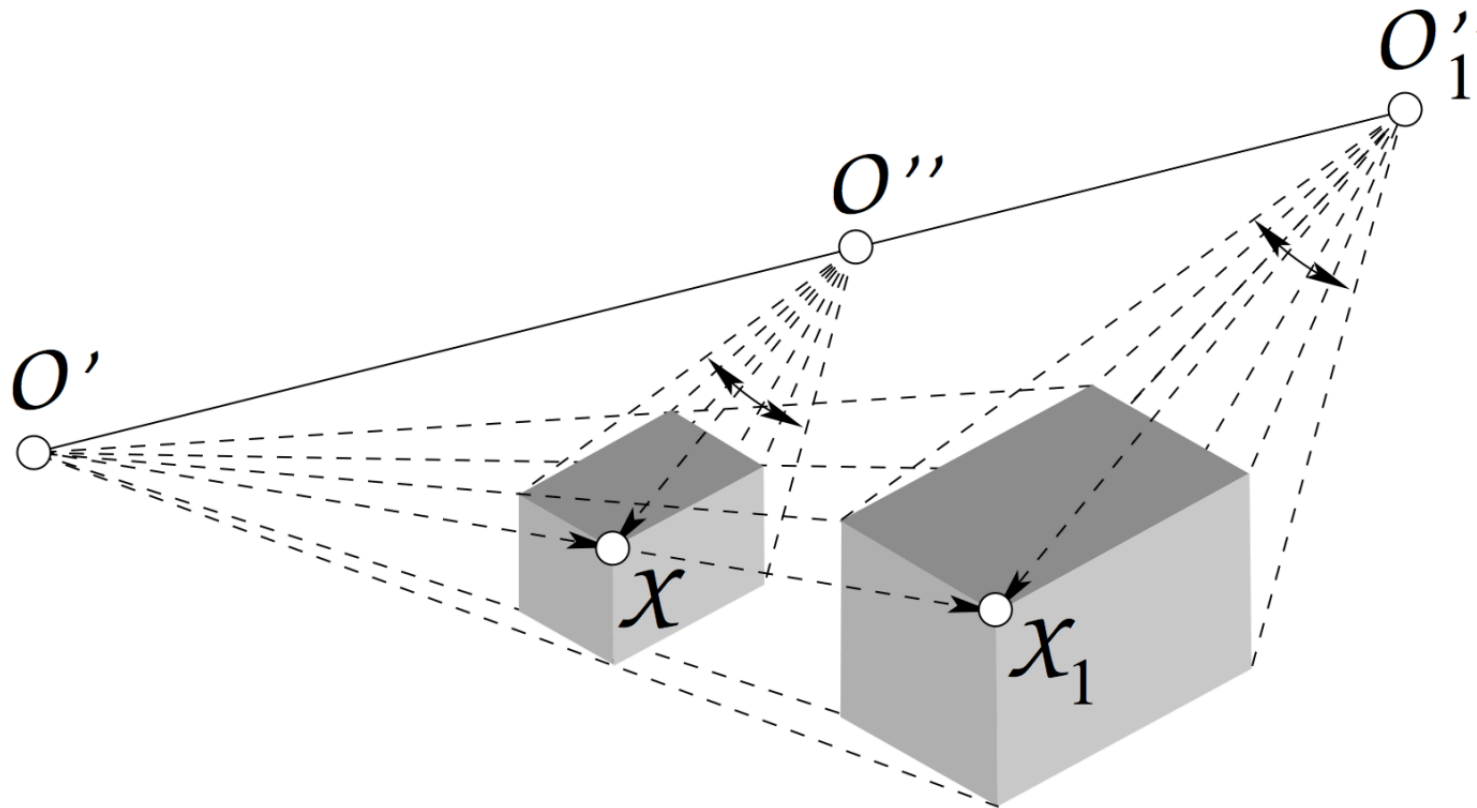
REMARK

Camera measure directions



- We **cannot** obtain:
 - the **global translation and rotation** (6 params)
 - the **scale** of the scene (1 param)

Camera measure directions



- We **can** obtain:
 - the **rotation of the second camera w.r.t. the first one** (3 params)
 - the **direction** of the line connecting the centers of projection (O' , O'') (2 params)

Photogrammetric model

- **photogrammetric model**: given two cameras images, we can reconstruct the object only **up to a similarity transform**
- **absolute orientation**: the **orientation of the photogrammetric model**
 - For obtaining the absolute orientation:
 - we need **at least 3 points** in 3D (to estimate the 7 parameters)
 - or **both camera locations**
 - or **location of one camera and the length of the baseline**
 - or **distances** on the real world
- In the following, we consider the photogrammetric model only

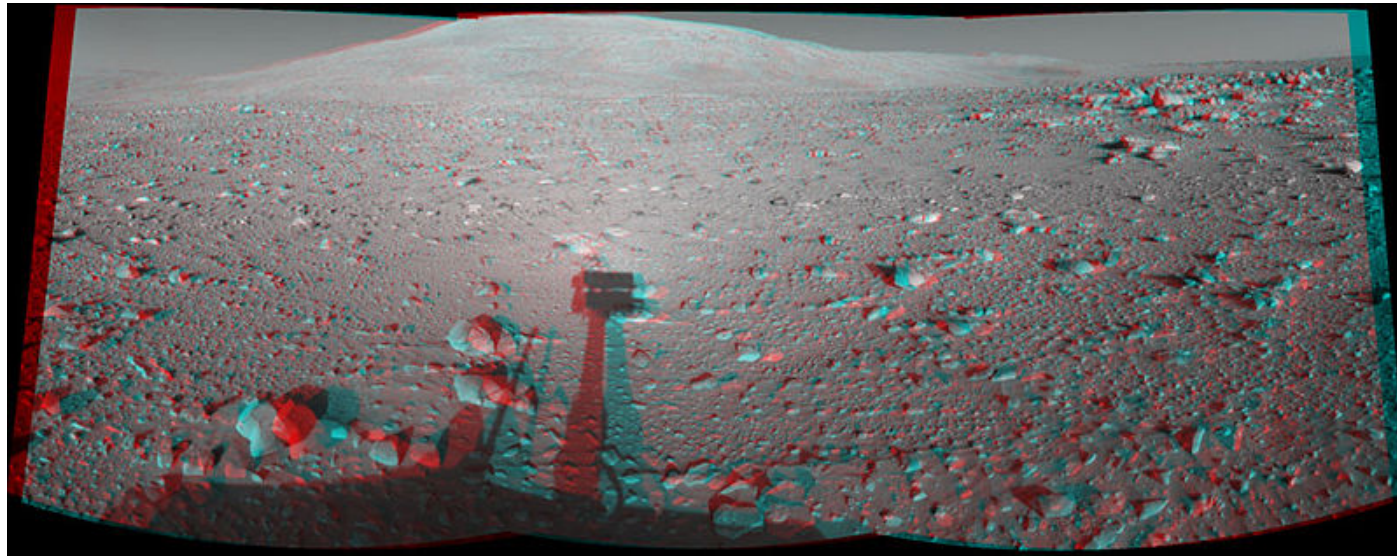
Application of stereo: Robotic exploration



Nomad robot searches for meteorites in Antarctica

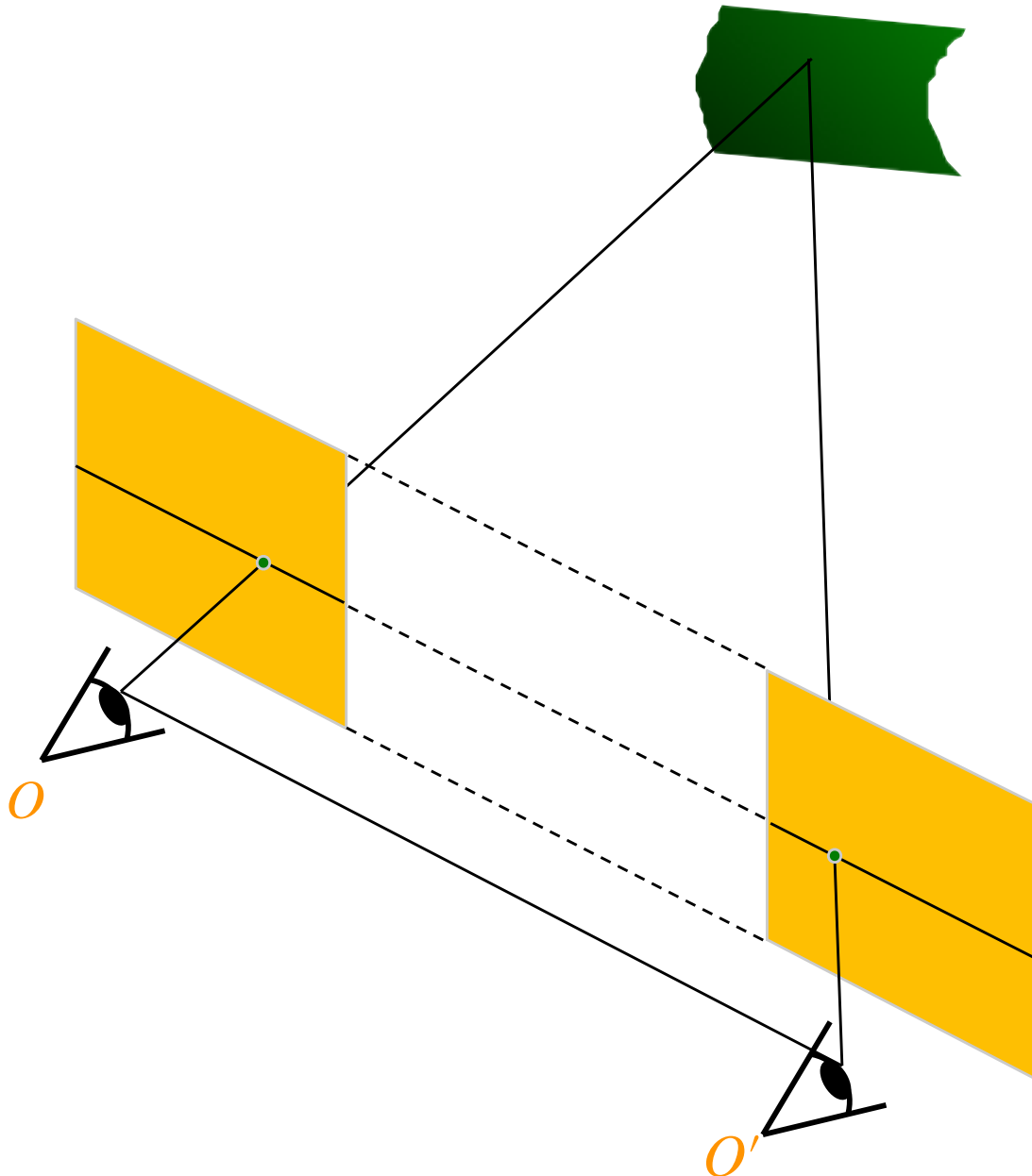


Real-time stereo on Mars



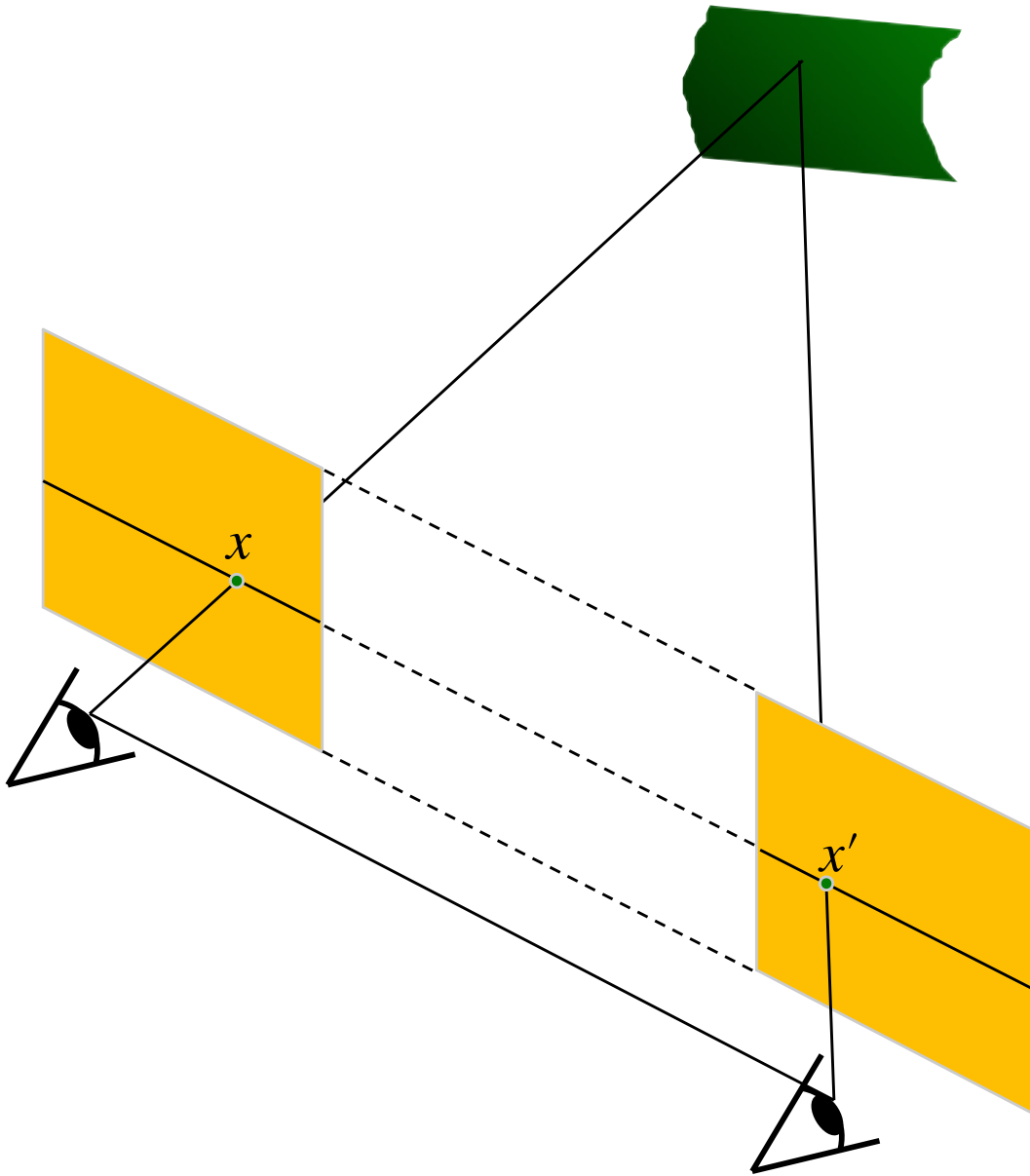
SIMPLE CASE: CO-PLANAR PLANES

Simplest Case: Parallel images



- **Image planes** of cameras are parallel to each other and to the baseline OO'
 - **Camera centers** are at same height
 - **Focal lengths** are the same
- ↓
- Coplanar plane images

Simplest Case: Parallel images

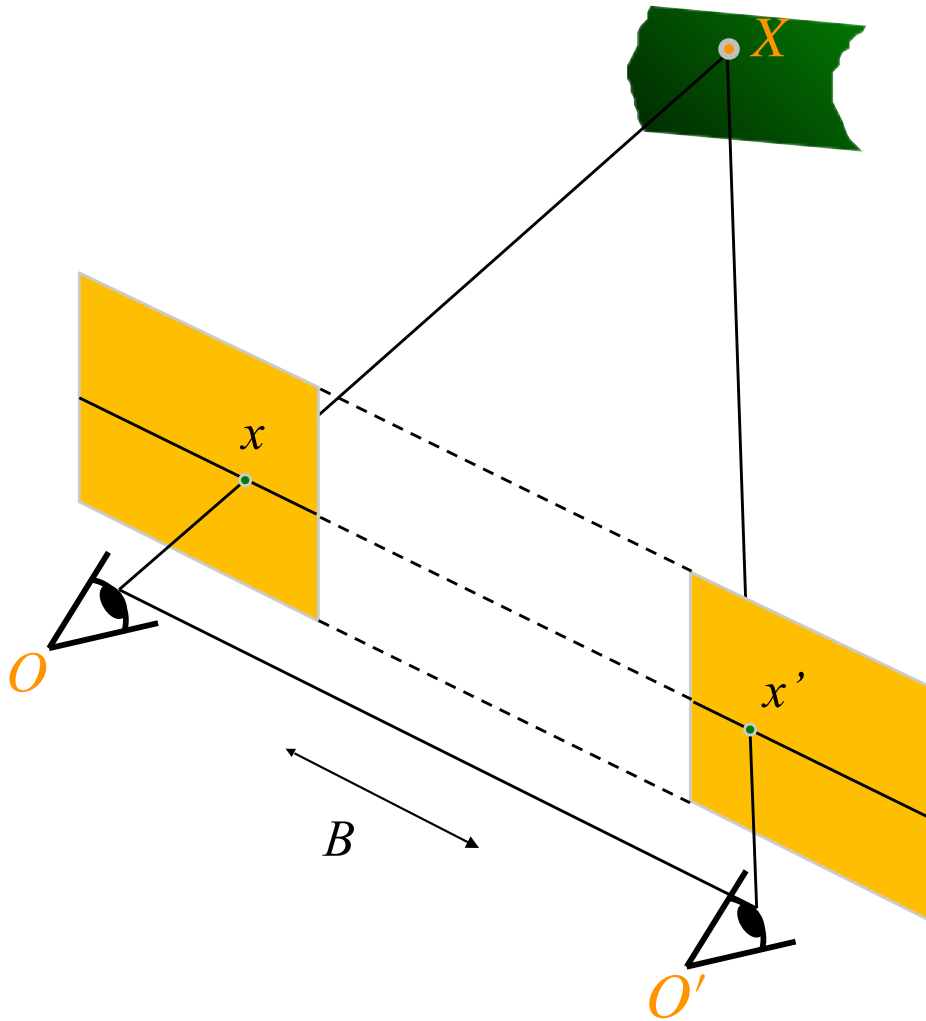


- Reliable matching:



given x find the best
matching x' at the same
height y on the other
image

Simplest Case: Parallel images



Disparity vs Depth and vs Baseline

- **Depth:** $Z = \frac{B \cdot f}{D}$
- **Disparity:** $D = \frac{f \cdot B}{Z}$
- $D \propto Z \implies \lim_{Z \rightarrow \infty} D = 0; \quad \lim_{Z \rightarrow 0} D = \infty$
 - a scene at infinity produce the same images on the two image planes (zero disparity)
 - as the object get closer and closer the disparity increases (difference in the projection)
- $D \propto B$
 - the most the camera are far away the disparity augment
 - we want to measure disparity precisely, we want the baseline large that gives more precise disparity measures

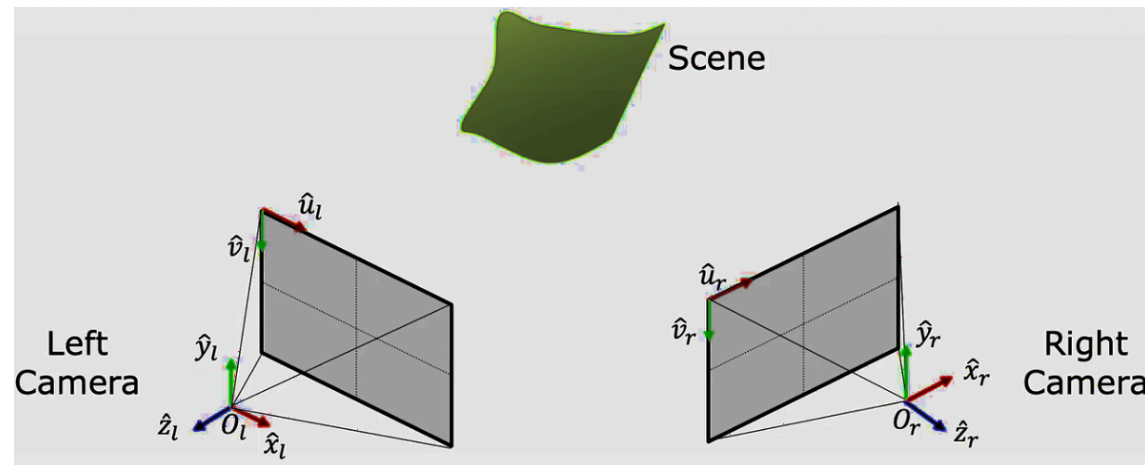
GENERAL CASE: UNCALIBRATED STEREO

Stereo vision: general statement

GOAL: given a calibrated binocular stereo pair (intrinsic parameters (f, c_x, c_y) are known), fuse it to produce a **depth map**

Problems to solve:

1. **Sparse Matching:** find a reliable matching between a few points u' in an image, u'' in the other one
2. Determine the **extrinsic parameters**, i.e relative position $\mathbf{t}$ and orientation R of cameras
3. Find **dense correspondence** (it's a 1D search)
4. Compute depth using triangulation

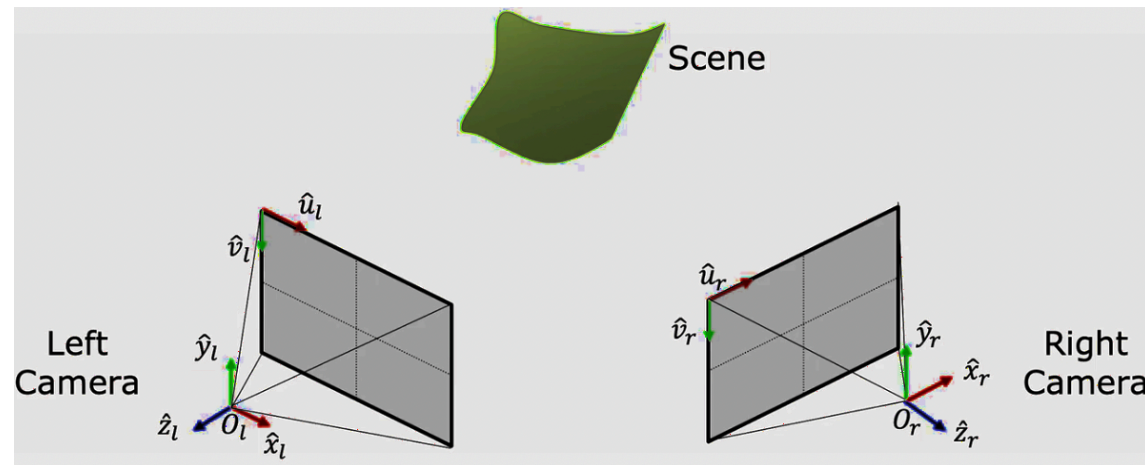


Stereo vision: general statement - Step 1

GOAL: given a calibrated binocular stereo pair (intrinsic parameters (f, c_x, c_y) are known), fuse it to produce a **depth map**

Problems to solve:

- 1. Sparse Matching:** find a reliable matching between a few points u' in an image, u'' in the other one
2. Determine the **extrinsic parameters**, i.e relative position t and orientation R of cameras
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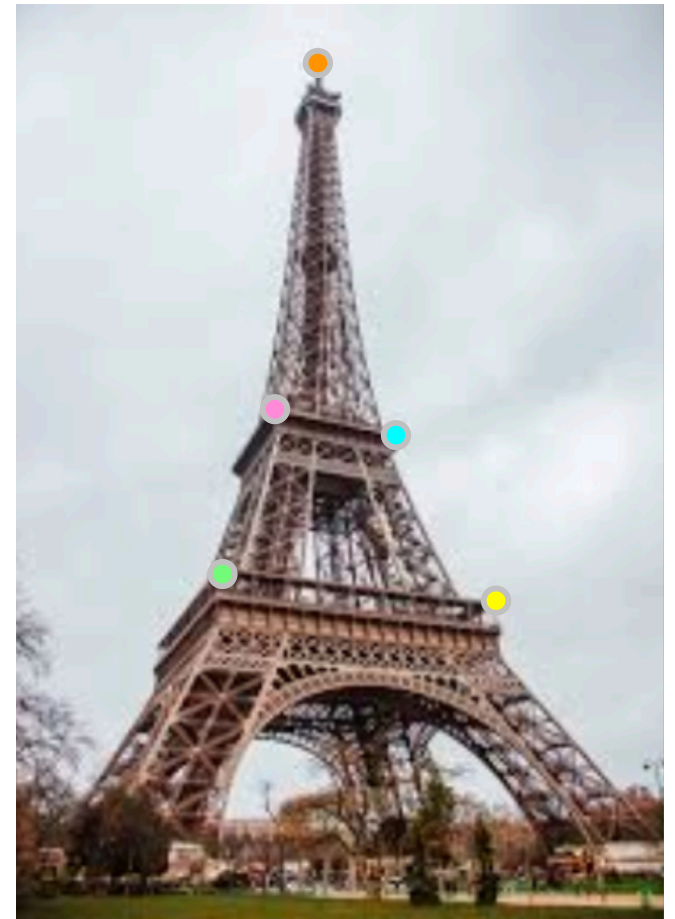
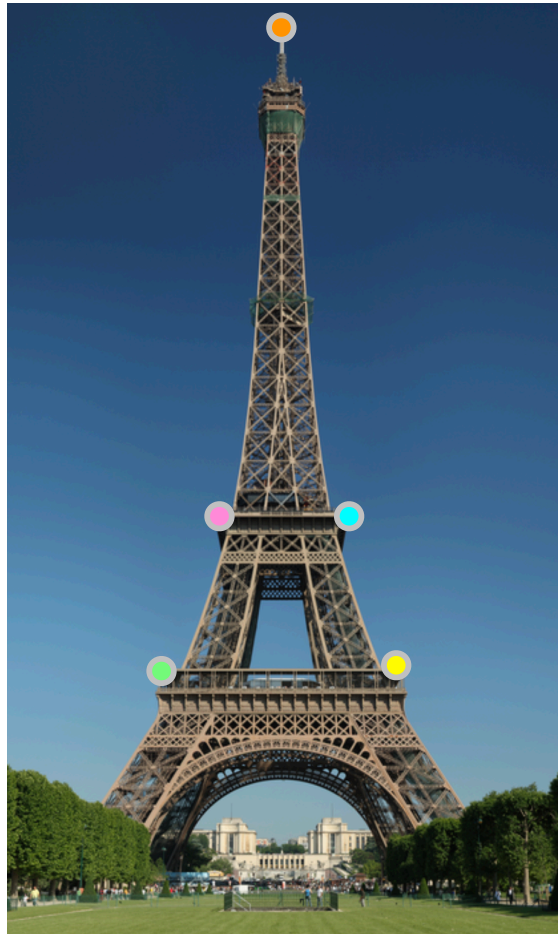


Stereo vision, uncalibrated stereo

- Steps:

- 1. **Sparse matching:** find a few reliable corresponding points

- manually
 - using SIFT

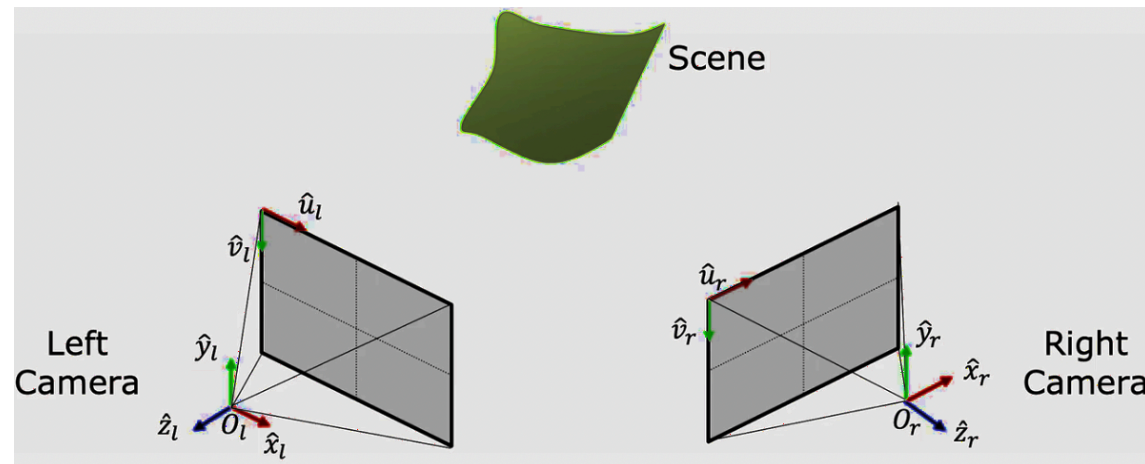


Stereo vision: general statement - Step 2

GOAL: given a calibrated binocular stereo pair (intrinsic parameters (f, c_x, c_y) are known), fuse it to produce a **depth map**

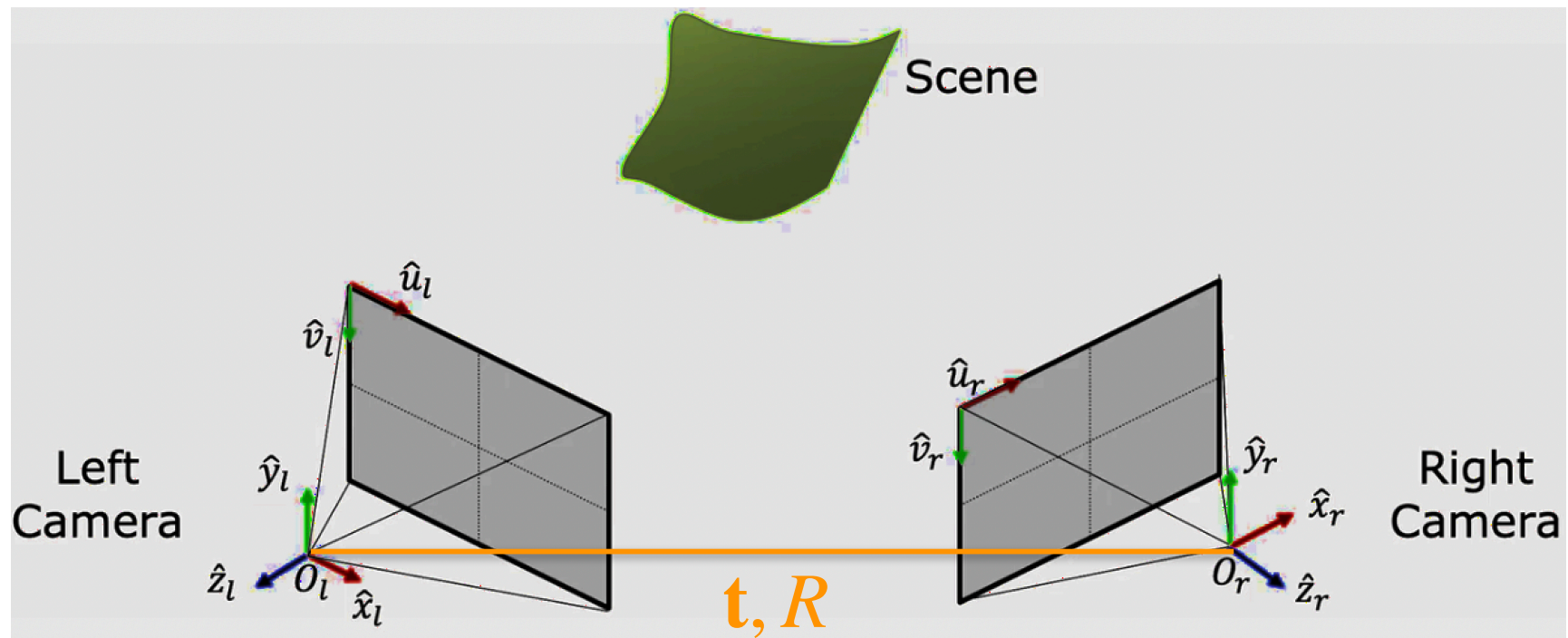
Problems to solve:

1. **Sparse Matching:** find a reliable matching between a few points u^l in an image, u^r in the other one
2. Determine the **extrinsic parameters**, i.e relative position t and orientation R of cameras
3. Find **dense correspondence** (it's a 1D search)
4. Compute depth using triangulation

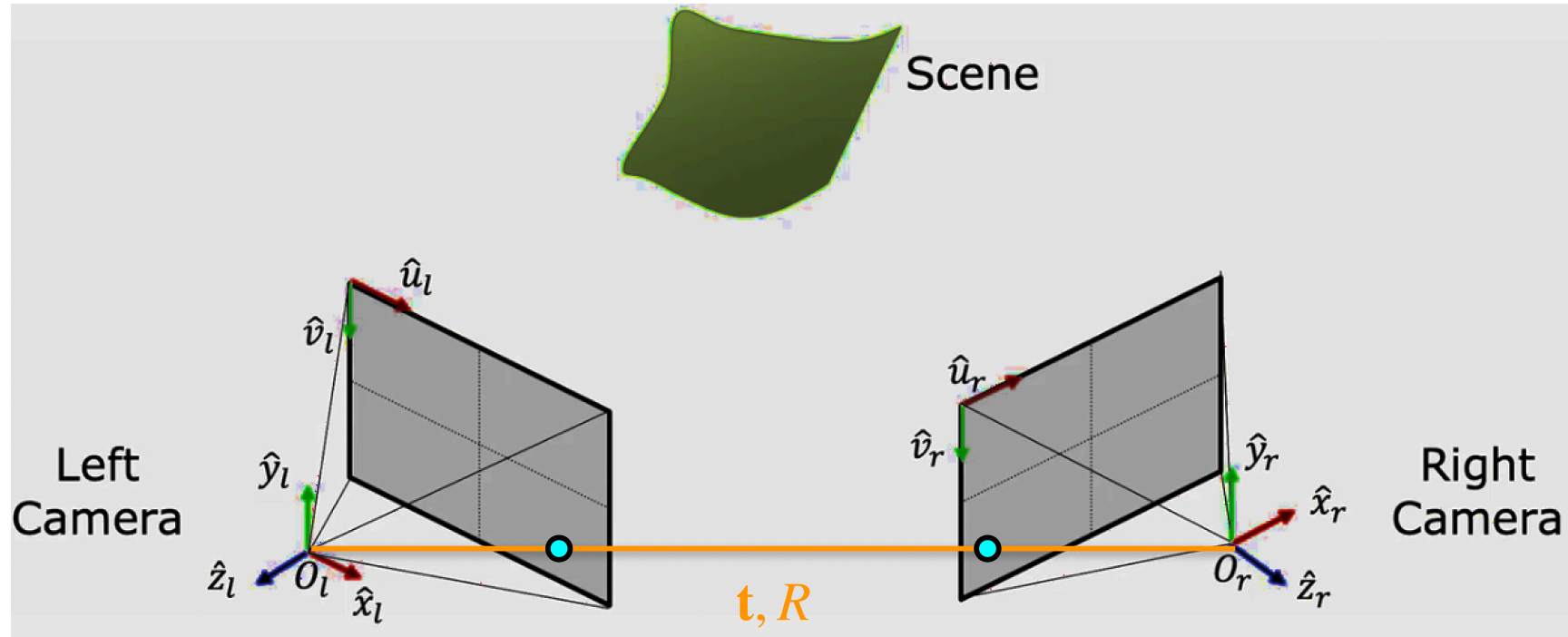


Stereo vision, uncalibrated stereo

- Steps:
 2. Determine the **relative position of the cameras $\mathbf{t}, R$**

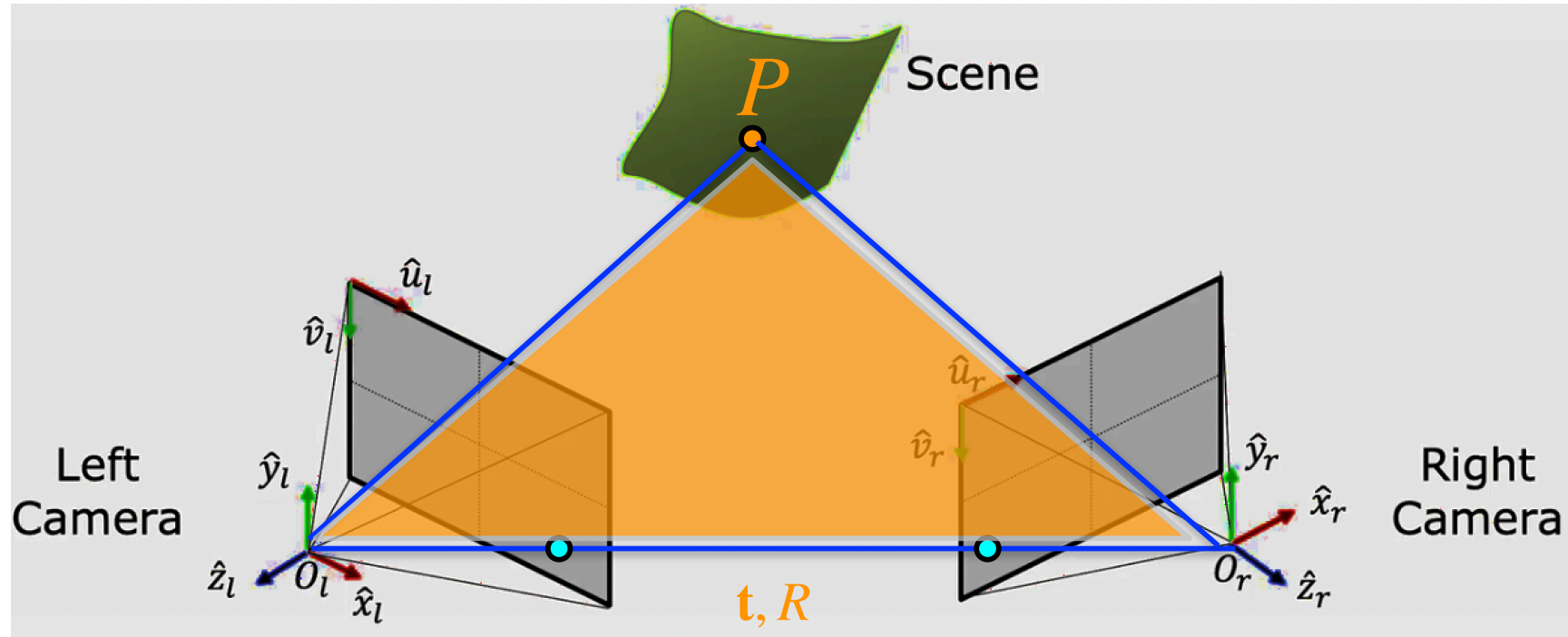


Epipolar geometry: epipoles



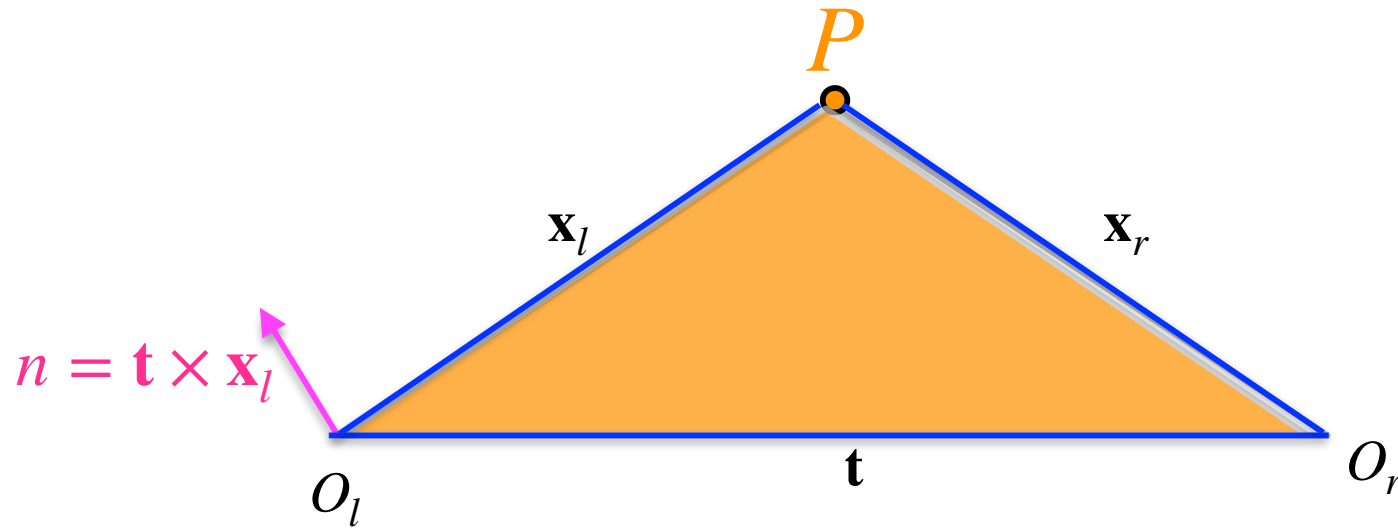
- **DEF: Epipoles:** intersection points between the baseline AND the image planes:
 - projections of each optic centre on the other image

Epipolar geometry: epipolar plane



- **DEF: Epipolar plane of scene point P :** plane passing through the camera centers AND point P

Epipolar geometry: epipolar constraint



- vector **normal** to the epipolar plane: $n = \mathbf{t} \times \mathbf{x}_l$
- it derives the **Epipolar constraint**: $\mathbf{n} \perp \mathbf{x}_l$

$$\mathbf{x}_l \cdot (\mathbf{t} \times \mathbf{x}_l) = 0$$

Epipolar geometry: epipolar constraint

1. Epipolar constraint $\mathbf{x}_l \cdot (\mathbf{t} \times \mathbf{x}_l) = 0$ using the $T_{\times}$ matrix:

$$\mathbf{x}_l \cdot (\mathbf{t} \times \mathbf{x}_l) = 0 \iff \mathbf{x}_l^T \cdot T_{\times} \cdot \mathbf{x}_l = 0 \quad \mathbf{t} = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} \rightarrow T_{\times} = \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \Rightarrow \mathbf{t} \times \mathbf{p} = T_{\times} \mathbf{p}$$

2. Roto-translation constraint:

$$\mathbf{x}_l = R\mathbf{x}_r + \mathbf{t}$$

3. Combine 1. and 2. we have:

$$\mathbf{x}_l^T \cdot T_{\times}(R \cdot \mathbf{x}_r + \mathbf{t}) = 0$$

$$\mathbf{x}_l^T \cdot (T_{\times}R\mathbf{x}_r + T_{\times}\mathbf{t}) = 0 \iff \mathbf{x}_l^T \cdot (T_{\times}R\mathbf{x}_r + \mathbf{t} \times \mathbf{t}) = 0 \iff \mathbf{x}_l^T T_{\times}R\mathbf{x}_r = 0$$

4. $E = T_{\times}R$ (Essential Matrix) $\rightarrow$ $\boxed{\mathbf{x}_l^T E \mathbf{x}_r = 0}$

[Longuet-Higgins 1981]

Epipolar geometry: epipolar constraint

$$E = T_{\times} R$$

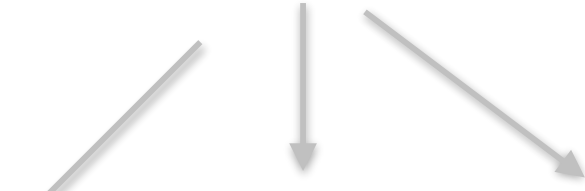
Remark: given E , we can **recover** $T_{\times}$ and R using **SVD**, since $T_{\times}$ is a **Skew-Symmetric** matrix ($a_{ij} = -a_{ji}$) and R is **orthonormal**

$$\begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} = \begin{bmatrix} 0 & -t_z & t_y \\ t_z & 0 & -t_x \\ -t_y & t_x & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

➔ ones calculated E , we decouple it and recover $\mathbf{t}$ and R using SVD

Epipolar geometry: calculate the essential matrix

- **Problem:** E relates the 3D position $\mathbf{x}_l$ of the scene point P wrt left camera to its position $\mathbf{x}_r$ wrt right camera

$$\mathbf{x}_l^T E \mathbf{x}_r = 0$$

$$[x_l \ y_l \ z_l] \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

- Unfortunately we don't have 3D coords $\mathbf{x}_r, \mathbf{x}_l$
- But we know the corresponding image coordinates
➔ **IDEA:** use the image coords to estimate E

Epipolar geometry: calculate the essential matrix

- Let's recall the **perspective projection equations** for the left camera:

$$u_l = f^{(l)} \frac{x_l}{z_l} + o_x^{(l)} \quad v_l = f^{(l)} \frac{y_l}{z_l} + o_y^{(l)} \quad \Longleftrightarrow \quad z_l u_l = f^{(l)} x_l + z_l o_x^{(l)} \quad z_l v_l = f^{(l)} y_l + z_l o_y^{(l)}$$

- Using homogeneous coords:

$$z_l \begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} = \begin{bmatrix} z_l u_l \\ z_l v_l \\ z_l \end{bmatrix} = \begin{bmatrix} f^{(l)} x_l + z_l o_x^{(l)} \\ f^{(l)} y_l + z_l o_y^{(l)} \\ z_l \end{bmatrix} = \boxed{\begin{bmatrix} f^{(l)} & 0 & o_x^{(l)} \\ 0 & f^{(l)} & o_y^{(l)} \\ 0 & 0 & 1 \end{bmatrix}} \begin{bmatrix} x_l \\ y_l \\ z_l \end{bmatrix}$$

- and inverting:

K_l , assumed to be known

$$\mathbf{x}_l^T = [u_l \quad v_l \quad 1] \cdot z_l \cdot K_l^{-1T}$$

- analogously for $\mathbf{x}_r$: $\mathbf{x}_r = K_r^{-1} \cdot z_r \cdot \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix}$

Epipolar geometry: Incorporating image coords

- Epipolar constraint using 3D points:

$$[x_l \ y_l \ z_l] \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

Product of 3
matrix 3×3
produce another
 3×3 matrix, that
we call F : the
fundamental
matrix

- Rewritten using the image coordinates:

$$[u_l \ v_l \ 1] \cdot z_l \cdot K_l^{-1T} \cdot \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \cdot K_r^{-1} \cdot z_r \cdot \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0$$

➡ z_l, z_r still unknown, but for sure $\neq 0$: we can eliminate them

Epipolar geometry: Incorporating image coords

- Epipolar constraint using 3D points:

$$[x_l \ y_l \ z_l] \begin{bmatrix} e_{11} & e_{12} & e_{13} \\ e_{21} & e_{22} & e_{23} \\ e_{31} & e_{32} & e_{33} \end{bmatrix} \begin{bmatrix} x_r \\ y_r \\ z_r \end{bmatrix} = 0$$

- Rewritten using the image coordinates:

$$[u_l \ v_l \ 1] \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r \\ v_r \\ 1 \end{bmatrix} = 0$$

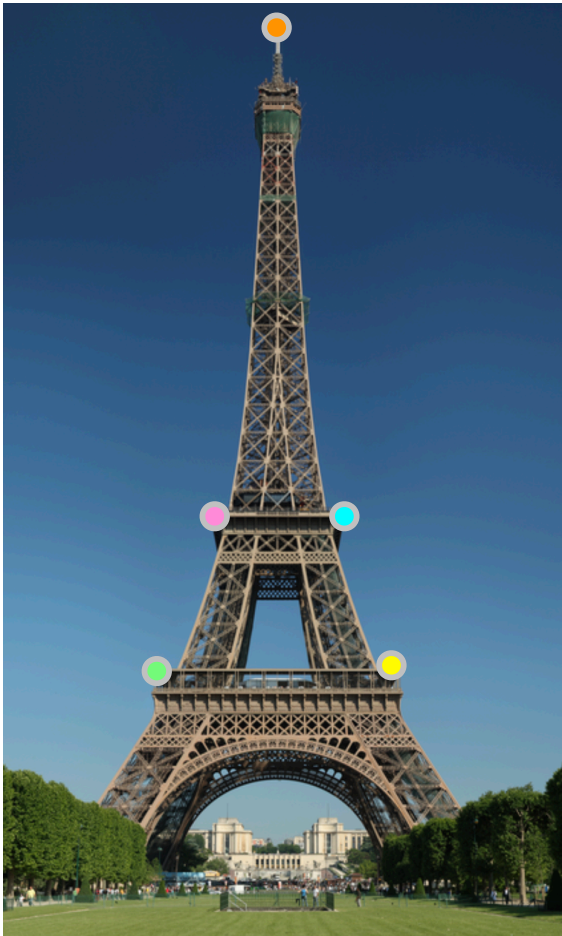
Fundamental Matrix F

$$F \rightarrow E = K_l^T F K_r \rightarrow E = T_{\times} R \rightarrow \boxed{\mathbf{t}, R}$$



Epipolar geometry: find the Fundamental Matrix

- ❖ Given a small set of corresponding features (Eg using SIFT):

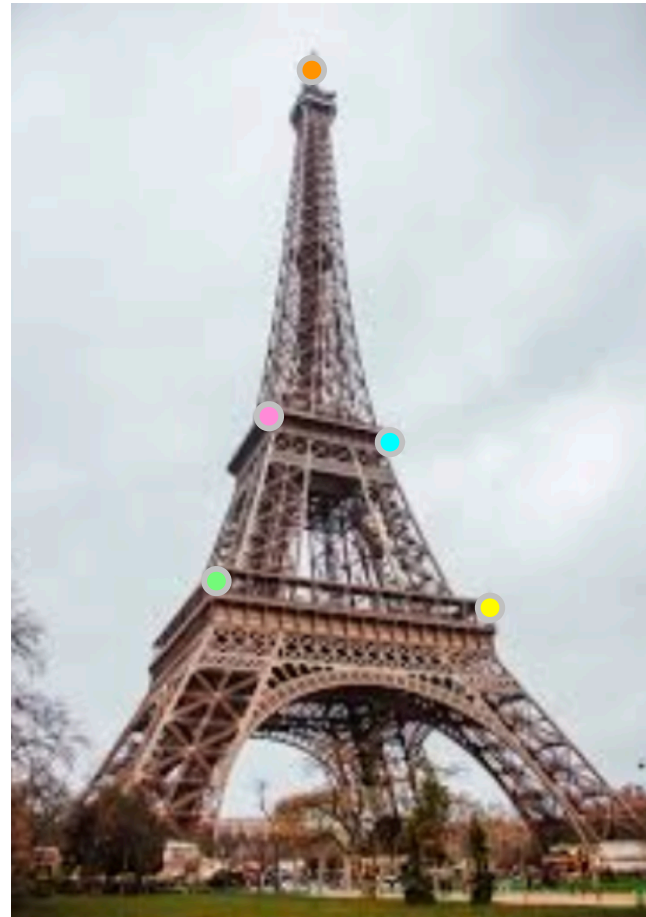


$$(u_l^{(1)}, v_l^{(1)})$$

$$(u_l^{(2)}, v_l^{(2)})$$

...

$$(u_l^{(m)}, v_l^{(m)})$$



$$(u_r^{(1)}, v_r^{(1)})$$

$$(u_r^{(2)}, v_r^{(2)})$$

...

$$(u_r^{(m)}, v_r^{(m)})$$

Epipolar geometry: find the Fundamental Matrix

- For each pair $\langle \mathbf{u}_l^{(i)}, \mathbf{u}_r^{(i)} \rangle$ write the epipolar constraint:

$$[u_l^{(i)} \quad v_l^{(i)} \quad 1] \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r^{(i)} \\ v_r^{(i)} \\ 1 \end{bmatrix} = 0$$

F unknown

- Expanding the matrix, we have the linear eq.:

$$(f_{11}u_r^{(i)} + f_{12}v_r^{(i)} + f_{13})u_l^{(i)} + (f_{21}u_r^{(i)} + f_{22}v_r^{(i)} + f_{23})v_l^{(i)} + f_{31}u_r^{(i)} + f_{32}v_r^{(i)} + f_{33} = 0$$

Epipolar geometry: find the Fundamental Matrix

- Rearrange in the form of a linear system:

$$\begin{bmatrix} u_l^{(1)}u_r^{(1)} & u_l^{(1)}v_r^{(1)} & u_l^{(1)} & v_l^{(1)}u_r^{(1)} & v_l^{(1)}v_r^{(1)} & v_l^{(1)} & u_r^{(1)} & v_r^{(1)} & 1 \\ \dots & & & & & & & & \\ u_l^{(i)}u_r^{(i)} & u_l^{(i)}v_r^{(i)} & u_l^{(i)} & v_l^{(i)}u_r^{(i)} & v_l^{(i)}v_r^{(i)} & v_l^{(i)} & u_r^{(i)} & v_r^{(i)} & 1 \\ \dots & & & & & & & & \\ u_l^{(m)}u_r^{(m)} & u_l^{(m)}v_r^{(m)} & u_l^{(m)} & v_l^{(m)}u_r^{(m)} & v_l^{(m)}v_r^{(m)} & v_l^{(m)} & u_r^{(m)} & v_r^{(m)} & 1 \end{bmatrix} \begin{bmatrix} f_{11} \\ f_{12} \\ f_{13} \\ f_{21} \\ f_{22} \\ f_{23} \\ f_{31} \\ f_{32} \\ f_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ \dots \\ 0 \\ \dots \\ 0 \end{bmatrix}$$

A , known

f , unknown

$$A\mathbf{f} = \mathbf{0}$$

Epipolar geometry: Fundamental matrix defined up to scale

REMARK: we are working in homogeneous coordinates, so that:

$$\begin{bmatrix} u_l^{(i)} & v_l^{(i)} & 1 \end{bmatrix} \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_r^{(i)} \\ u_r^{(i)} \\ 1 \end{bmatrix} = 0 = \begin{bmatrix} u_l^{(i)} & v_l^{(i)} & 1 \end{bmatrix} \begin{bmatrix} kf_{11} & kf_{12} & kf_{13} \\ kf_{21} & kf_{22} & kf_{23} \\ kf_{31} & kf_{32} & kf_{33} \end{bmatrix} \begin{bmatrix} u_r^{(i)} \\ u_r^{(i)} \\ 1 \end{bmatrix}$$

- F and kF describe the same epipolar geometry
- F is defined up to a scale
- We choose the scale s.t. $\|f\|^2 = 1$

Epipolar geometry: Fundamental matrix defined up to scale

- Find F :
 1. Solve the constrained linear least squares problem:

$$\min_{\mathbf{f}} \|A\mathbf{f}\|^2 \quad s.t. \quad \|\mathbf{f}\|^2 = 1$$

[solution: the eigenvector of the matrix $A^T A$ corresponding to the smallest eigenvalue, normalized so that its norm is 1]

2. Rearrange $\mathbf{f}$ to obtain the **fundamental matrix** F

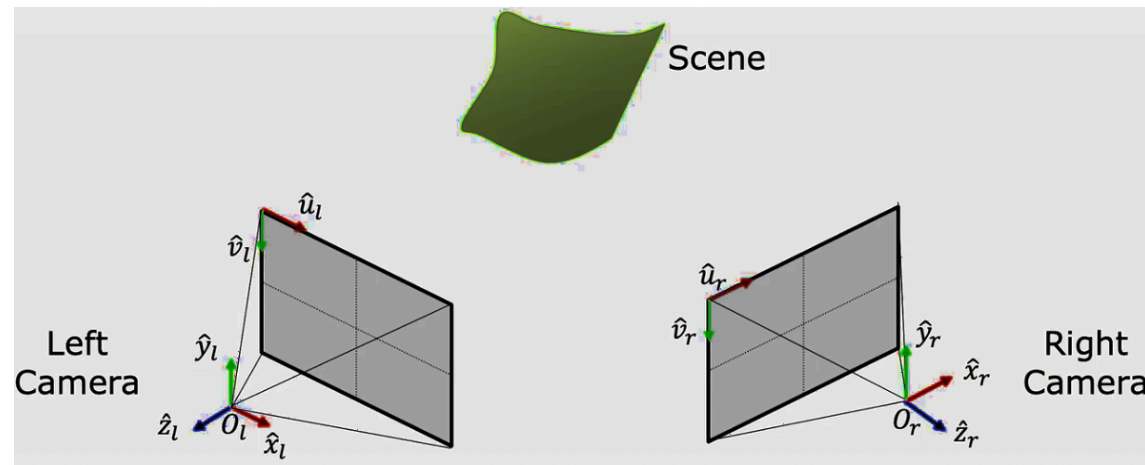
- Derive the **essential matrix** $E = K_l^T F K_r$
- Finally **extract** $T_{\times}$, and R using the SVD: $E = T_{\times} R$

Stereo vision: general statement

GOAL: given a calibrated binocular stereo pair (intrinsic parameters (f, c_x, c_y) are known), fuse it to produce a **depth map**

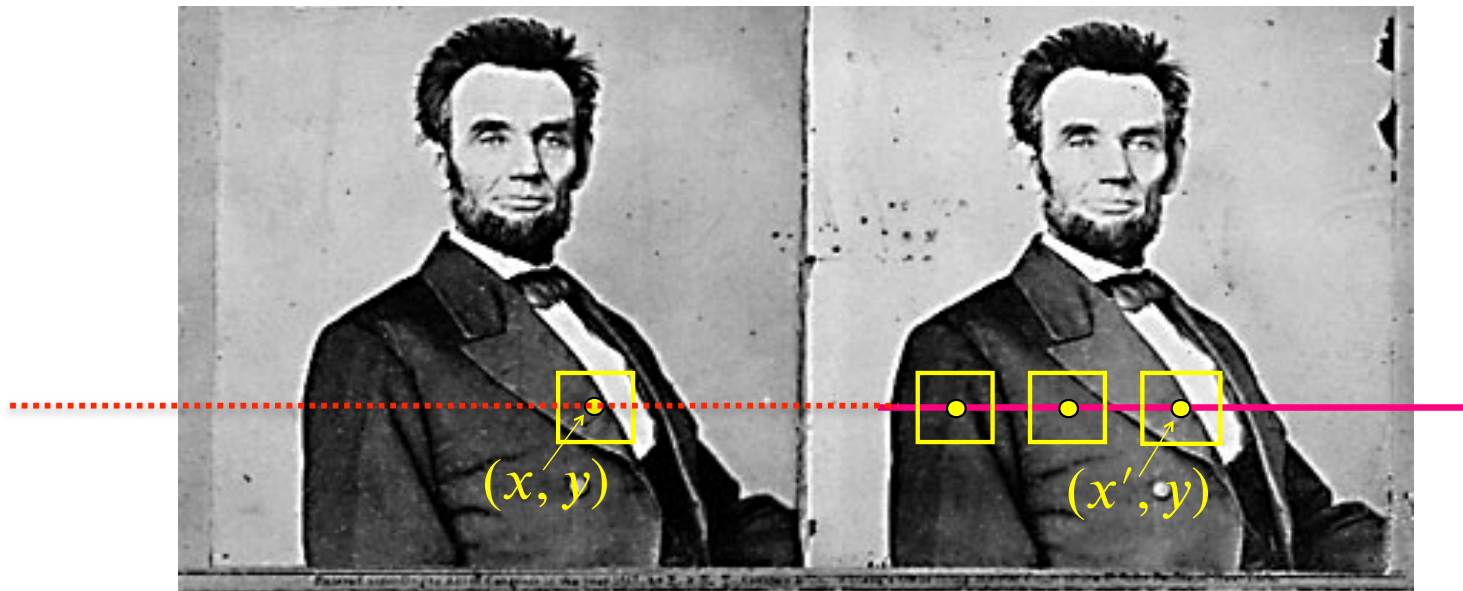
Problems to solve:

1. **Sparse Matching:** find a reliable matching between a few points $\mathbf{u}'$ in an image, $\mathbf{u}''$ in the other one
2. Determine the **extrinsic parameters**, i.e relative position $\mathbf{t}$ and orientation R of cameras
3. Find **dense correspondence** (it's a 1D search)
4. Compute depth using triangulation

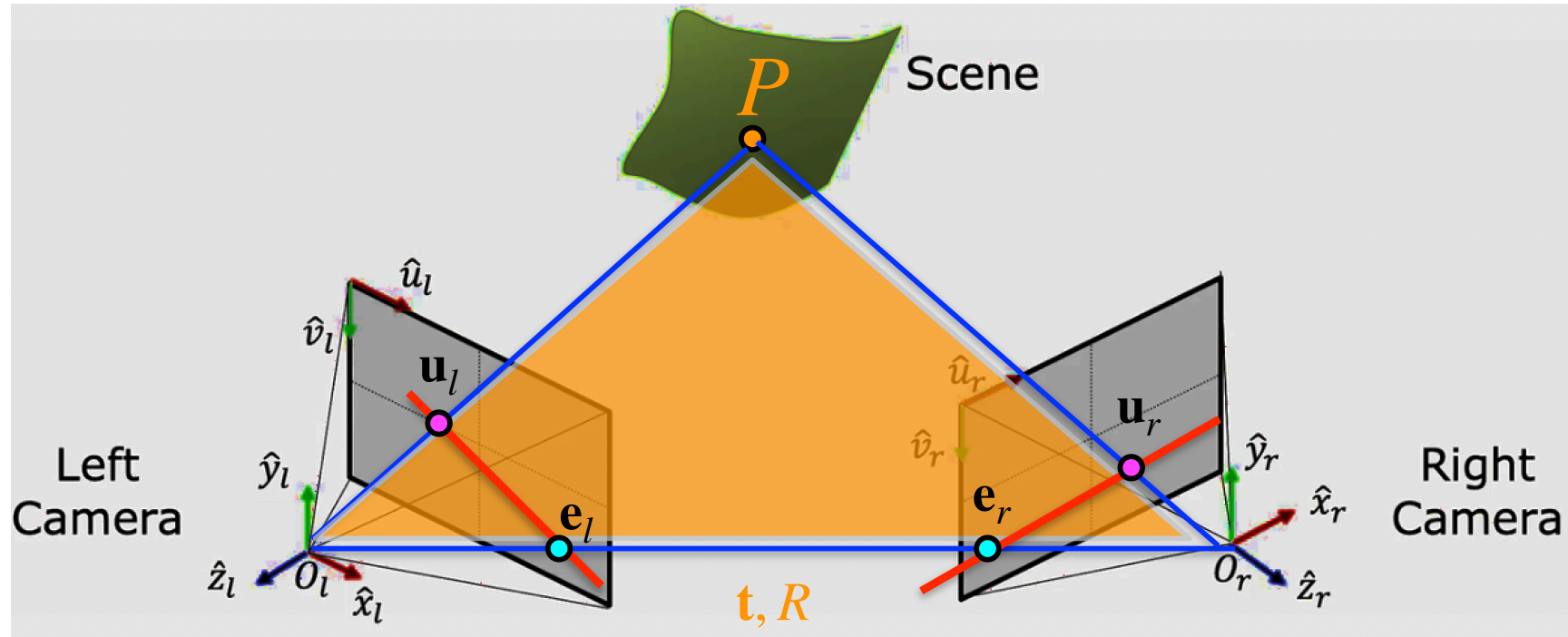


Finding dense correspondences: simple case

- Step 3: find dense correspondence (it's a 1D search)
- In case of co-planar images, corresponding points lie on the same horizontal scan-line
- Finding correspondence is a 1D match



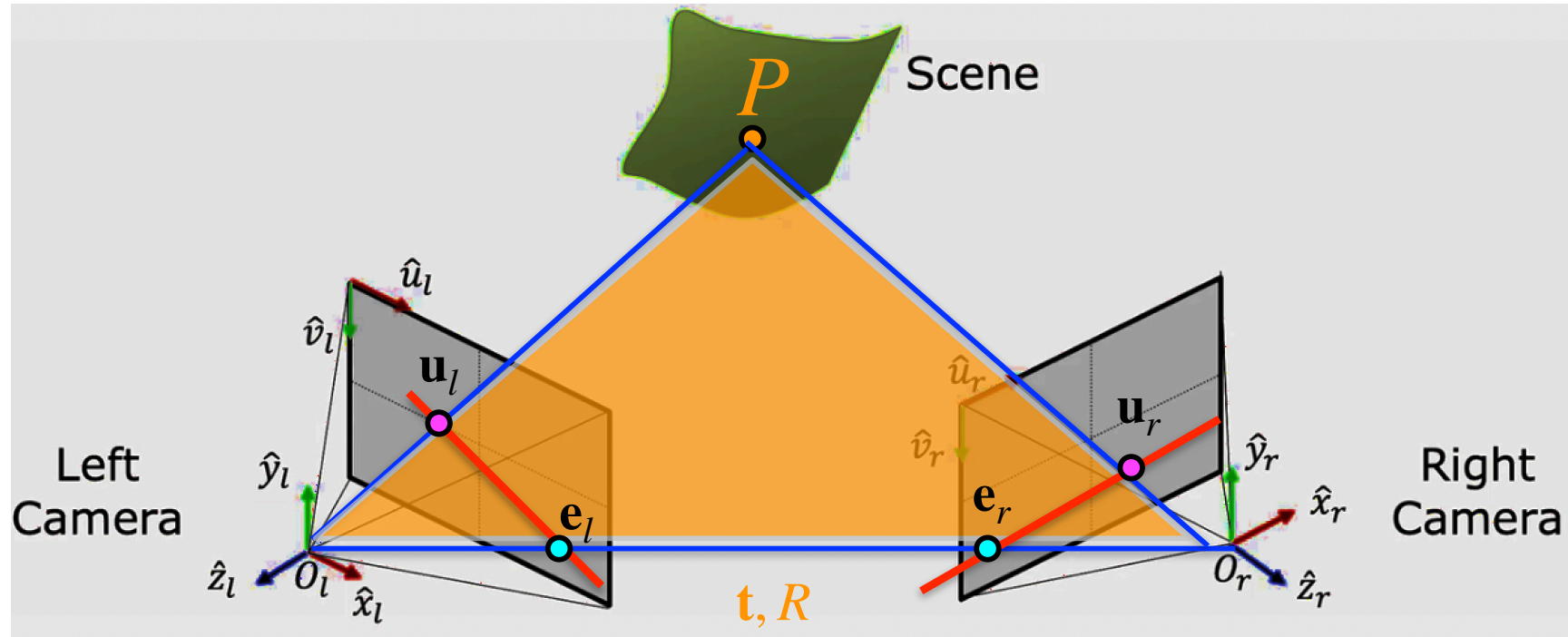
Epipolar geometry: epipolar lines



- **Epipolar lines**

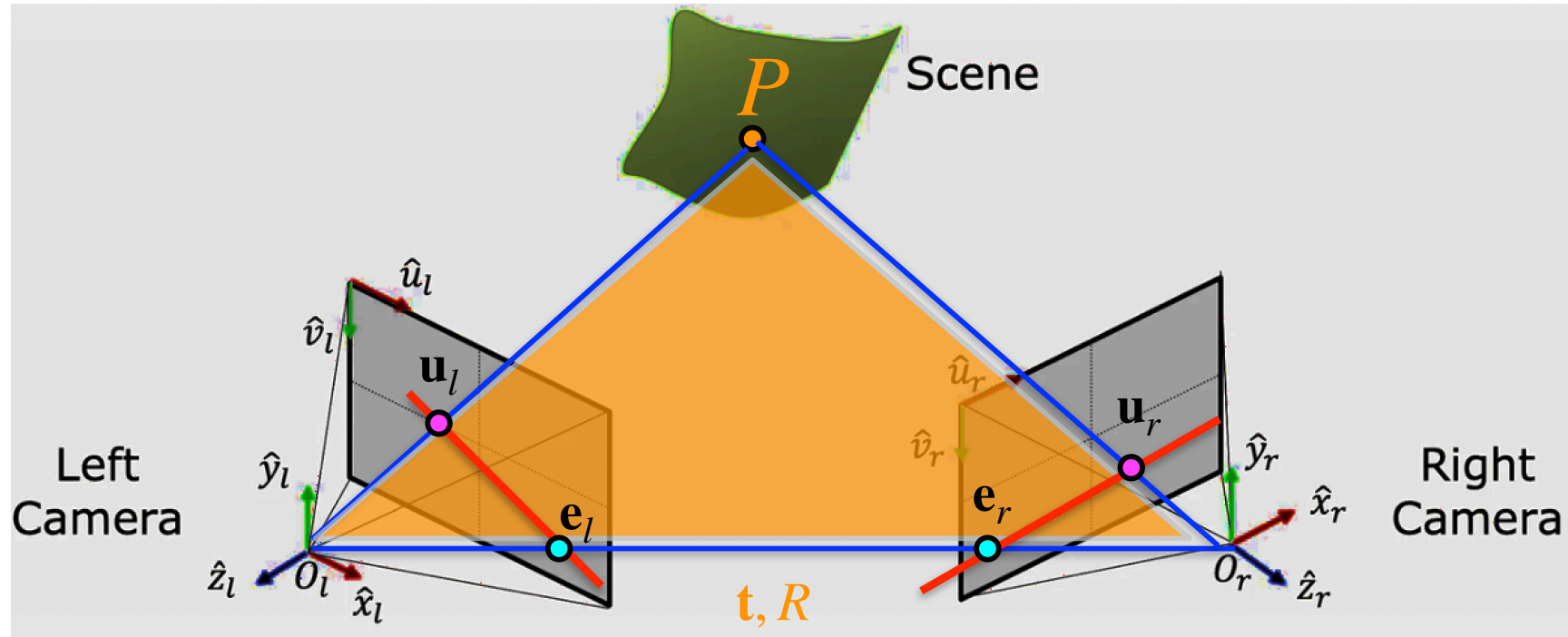
Intersection between the image plane and the epipolar plane (pass through the **epipoles** and the **image point**)

Epipolar geometry: epipolar lines



- Every scene point P has two corresponding **epipolar lines**, one per image plane

Epipolar geometry: epipolar plane



- **REMARK:** Given a point in one image $\mathbf{u}_l$, the corresponding point on the other image $\mathbf{u}_r$ must lie on the epipolar line
- Given a point $\mathbf{u}_l$ its matching has to be looked for with 1D search on the corresponding epipolar line on the other image

Epipolar geometry: finding epipolar line

- ➔ **GOAL**: find the epipolar line on the right image, having:
- Fundamental Matrix F
 - the image point $\mathbf{u}_l$ on the left image

- Remember it holds the epipolar constraint:

$$[u_r \quad v_r \quad 1] \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} u_l \\ v_l \\ 1 \end{bmatrix} = 0$$

- Expanding it we have:

$$(f_{11}u_l + f_{12}v_l + f_{13})u_r + (f_{21}u_l + f_{22}v_l + f_{23})v_r + (f_{31}u_l + f_{32}v_l + f_{33}) = 0$$

- simplified to express **the right epipolar line** (similarly for the left epipolar line):

$$a_l \cdot u_r + b_l \cdot v_r + c_l = 0$$

Epipolar geometry: finding epipolar plane, an example

- Given the Fundamental matrix

$$F = \begin{bmatrix} -.003 & -.028 & 13.19 \\ -.003 & -.008 & -29.2 \\ 2.97 & 56.38 & -9999 \end{bmatrix}$$

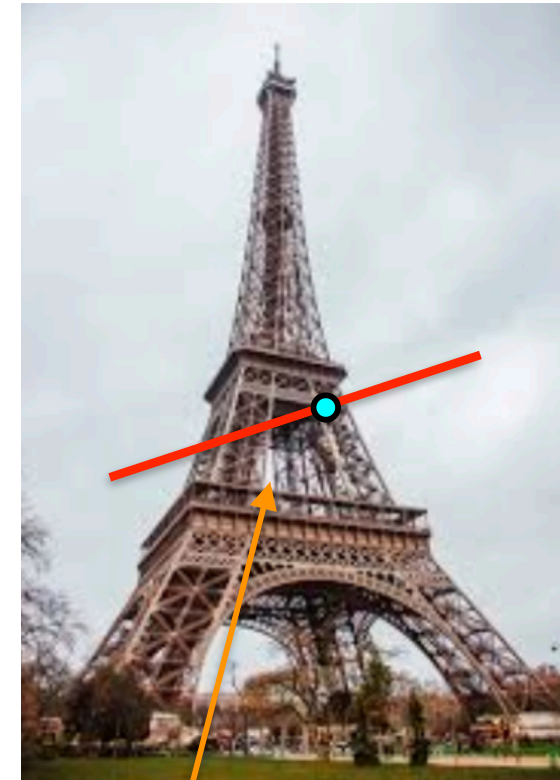
- and the left image point:

$$\tilde{\mathbf{u}}_l = \begin{bmatrix} 343 \\ 221 \\ 1 \end{bmatrix},$$

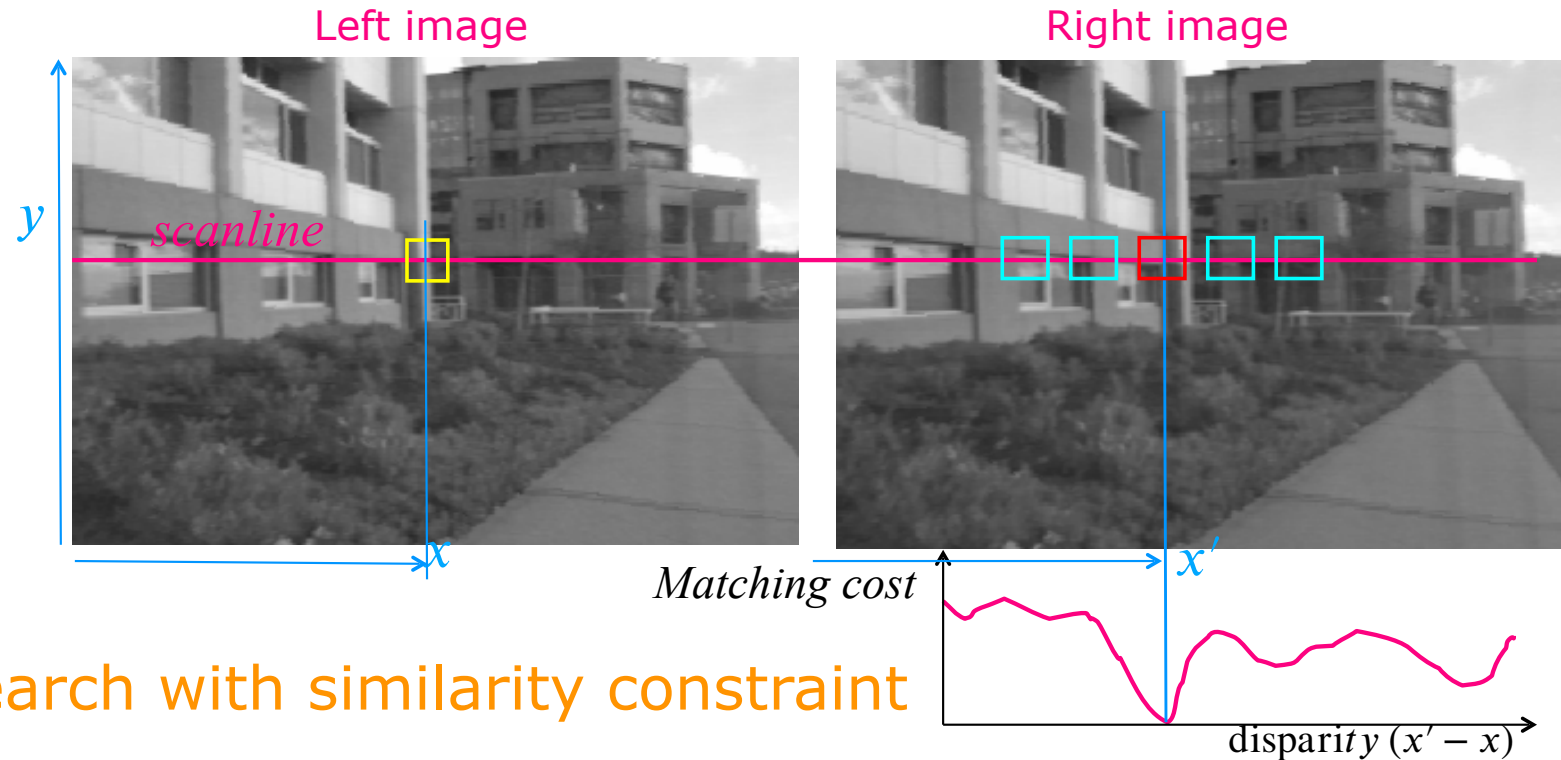
- Epipolar line in the right image:

$$\begin{bmatrix} u_r & v_r & 1 \end{bmatrix} \begin{bmatrix} -.003 & -.028 & 13.19 \\ -.003 & -.008 & -29.2 \\ 2.97 & 56.38 & -9999 \end{bmatrix} \begin{bmatrix} 343 \\ 221 \\ 1 \end{bmatrix} = 0 \quad \Longleftrightarrow \quad .03u_r + .99v_r = 265$$

- Find best match along the epipolar line



Matching algorithm – local approach – Correlation



Correspondence search with similarity constraint

1. Slide a window $W_L(x, y)$ taken on I_{Left} on the scanline in I_{Right}
2. Find the window $W_R(x', y)$ that minimizes a matching cost

Matching cost:

- **Normalized Correlation**
- **SSD** – Sum of Squared Differences
- **SAD** – Sum of Absolute Differences

Matching algorithm – local approach – Normalized Correlation

Correlation between windows with Mean 0 and Std 1

Robust to illumination variations

Left image



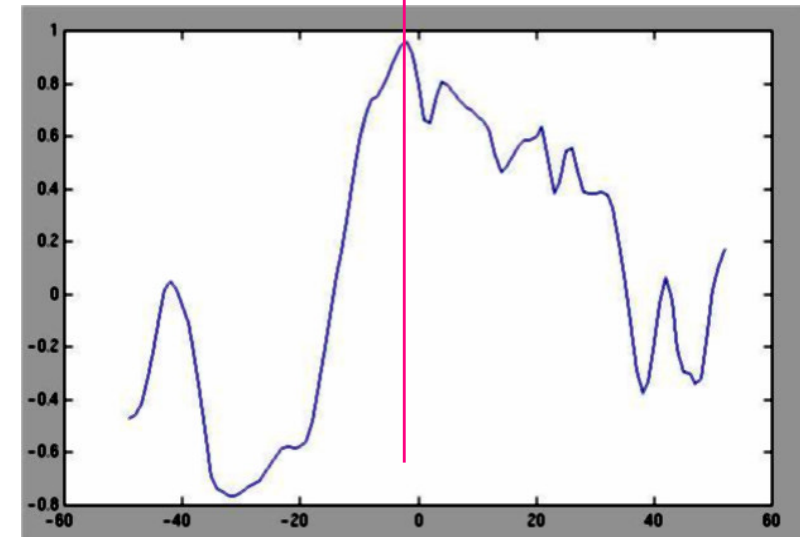
Right image



$$I_L(x+i, y+j) \rightarrow w_L(k), k \in W$$

$$I_R(x'+i, y+j) \rightarrow w_R(k), k \in W$$

$$C(x, y, d) = \frac{\sum_{k \in W} \left[\left(w_L(k) - \overline{w_L} \right) \cdot \left(w_R(k) - \overline{w_R} \right) \right]}{\left\| \mathbf{w}_L - \overline{w_L} \right\| \left\| \mathbf{w}_R - \overline{w_R} \right\|}$$



Matching algorithm – local approach – SSD SAD



SSD – *Sum of Squared Differences*:

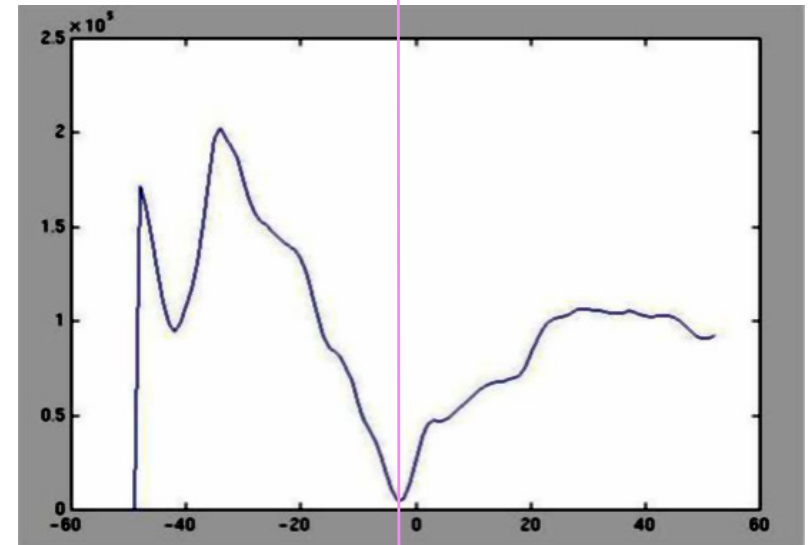
$$I_L(x+i, y+j) \rightarrow w_L(k), k \in W$$

$$I_R(x'+i, y+j) \rightarrow w_R(k), k \in W$$

$$C(x, y, d) = \sum_{k \in W} (w_L(k) - w_R(k))^2, \quad d = x - x'$$

SAD – *Sum of Absolute Differences*:

$$C(x, y, d) = \sum_{k \in W} |w_L(k) - w_R(k)|$$



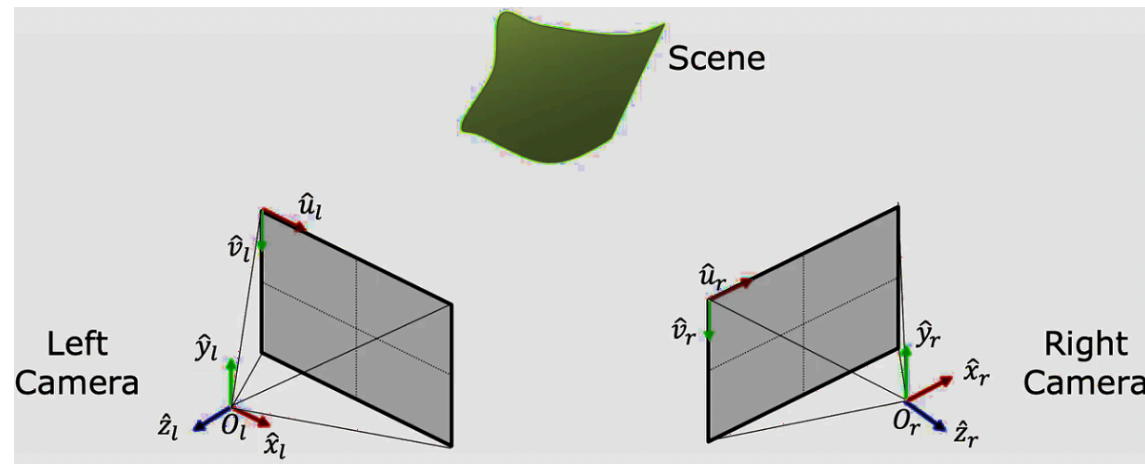
SSD($x - x'$)

Stereo vision: general statement

GOAL: given a calibrated binocular stereo pair (intrinsic parameters (f, c_x, c_y) are known), fuse it to produce a **depth map**

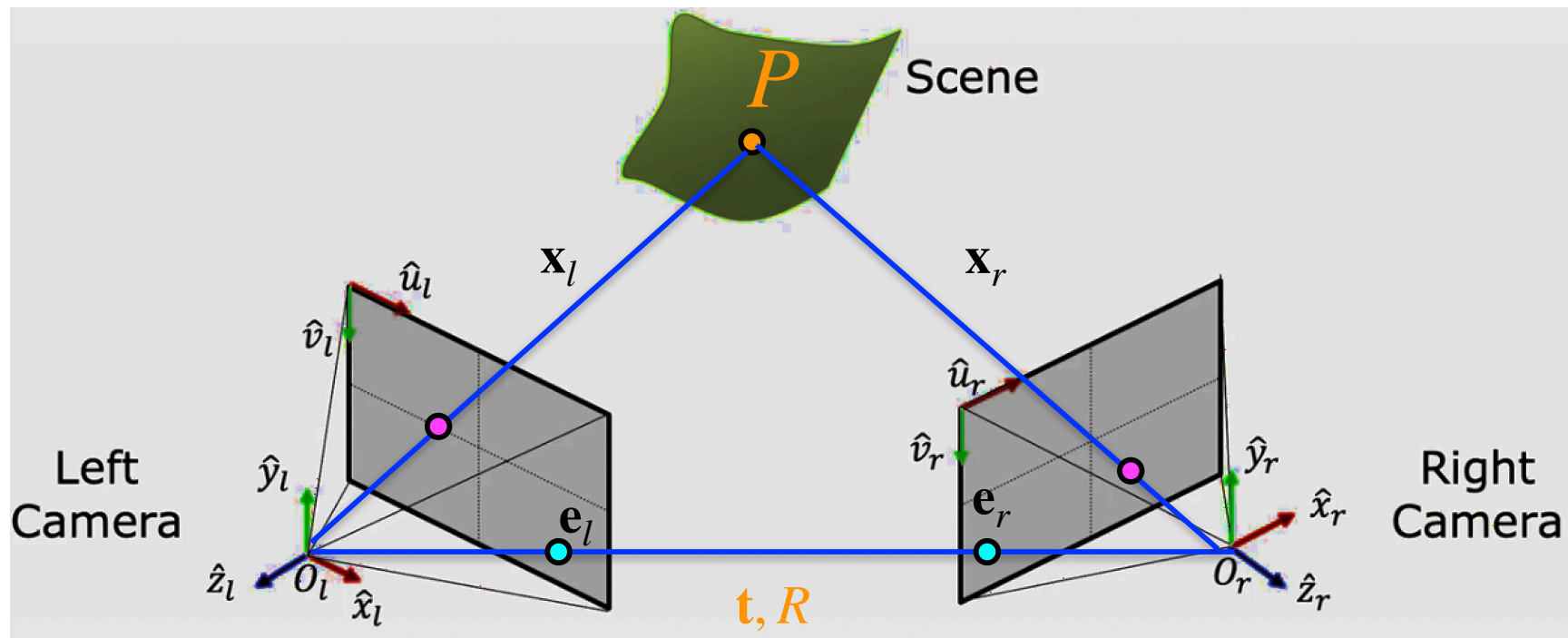
Problems to solve:

1. **Sparse Matching:** find a reliable matching between a few points $\mathbf{u}'$ in an image, $\mathbf{u}''$ in the other one
2. Determine the **extrinsic parameters**, i.e relative position $\mathbf{t}$ and orientation R of cameras
3. Find **dense correspondence** (it's a 1D search)
4. Compute depth using triangulation



Computing depth

- Step 4: find the depth of the point, i.e z_l or z_r



Triangulation: linear algebraic approach

- Having calibrated the camera we know M_l, M_r such that:

- $\mathbf{p}_l = M_l \cdot \mathbf{P}, \quad \mathbf{p}_r = M_r \cdot \mathbf{P}$

- it holds null cross product:

- $\mathbf{p}_l \times M_l \cdot P = 0, \quad \mathbf{p}_r \times M_r \cdot P = 0$

- arranging in a linear system:

$$\begin{bmatrix} \mathbf{p}_l \times M_l \mathbf{P} \\ \mathbf{p}_r \times M_r \mathbf{P} \end{bmatrix} = \begin{bmatrix} \mathbf{p}_l \times M_l \mathbf{P} \\ \mathbf{p}_r \times M_r \mathbf{P} \end{bmatrix} = \begin{bmatrix} \mathbf{p}_l \times M_l \\ \mathbf{p}_r \times M_r \end{bmatrix} \mathbf{P} = A' \begin{bmatrix} X_P \\ Y_P \\ Z_P \\ 1 \end{bmatrix} = [\mathbf{0}]$$

- rearranging so that $A_{[6 \times 3]} = A'[:, 0 : 3]; \quad \mathbf{b} = A'[:, 3]:$

$$A_{[6 \times 3]} \cdot \begin{bmatrix} X_P \\ Y_P \\ Z_P \end{bmatrix} = \mathbf{b}_{[6 \times 1]} \iff P = \begin{bmatrix} X_P \\ Y_P \\ Z_P \end{bmatrix} = (A^T A)^{-1} A^T \mathbf{b}$$

Least squares sol.

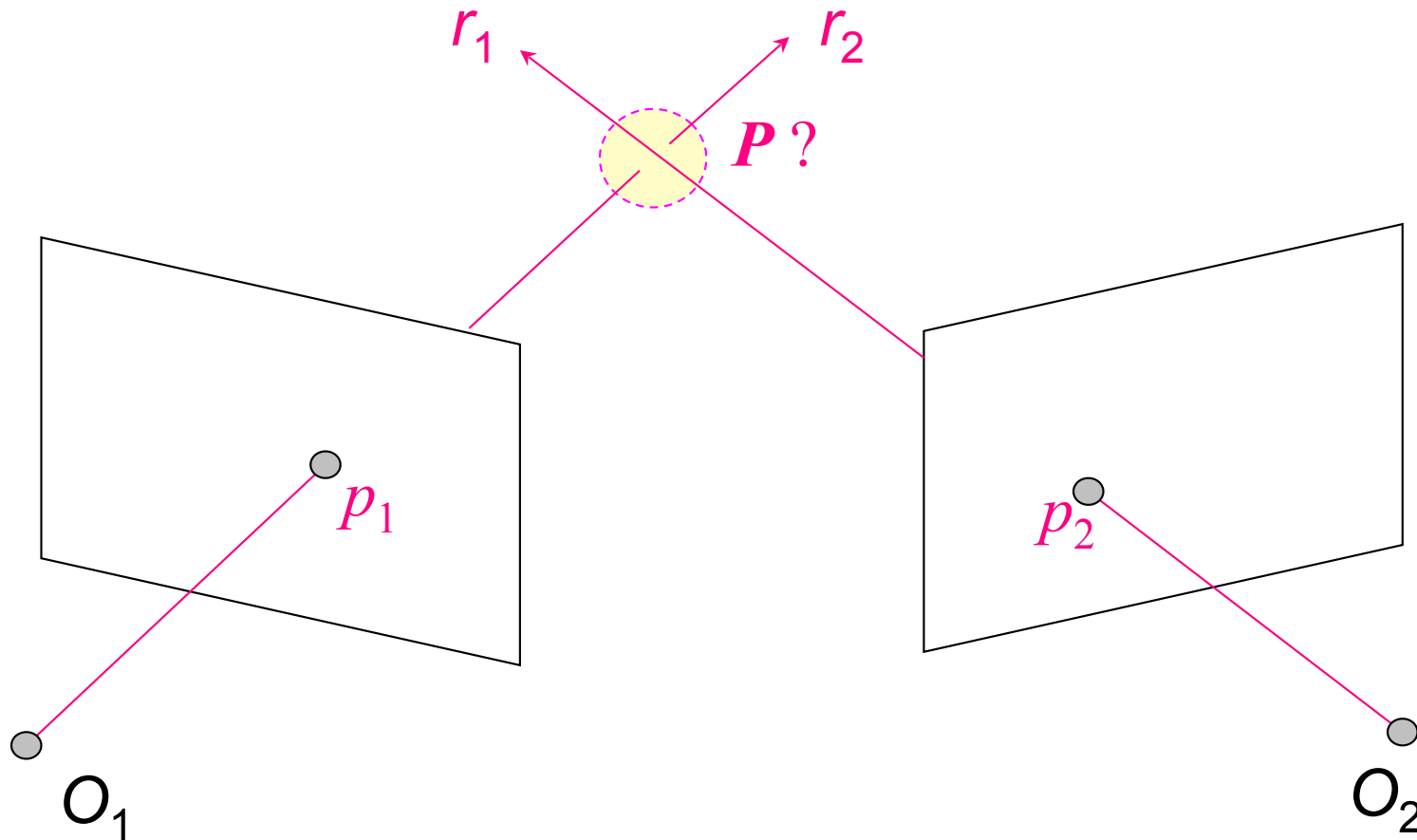
To be repeated
for every point tablet

Triangulation

Problem in real scenarios

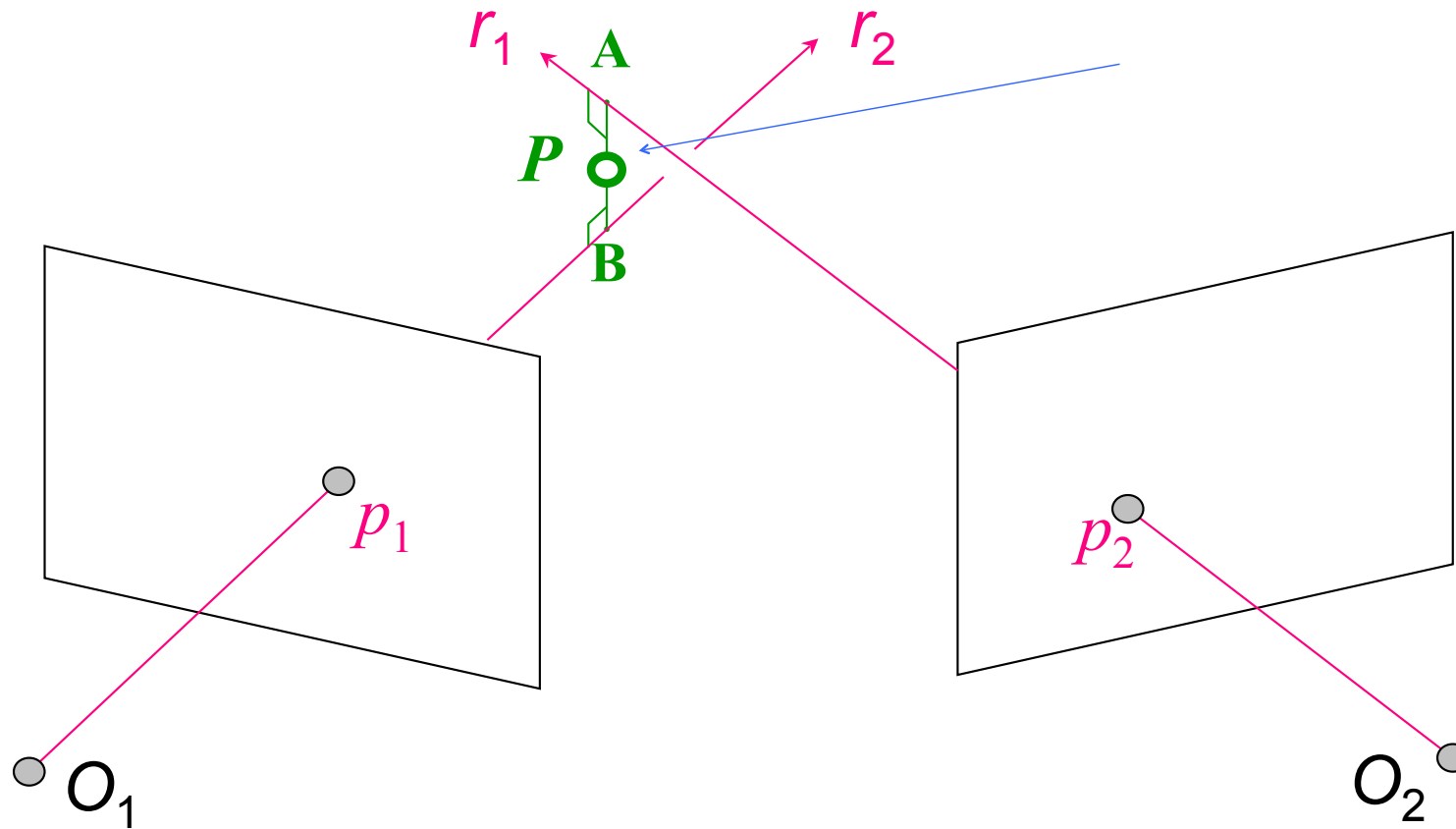
The rays r_1 and r_2 do not intersect each other

- Because of localization/calibration errors



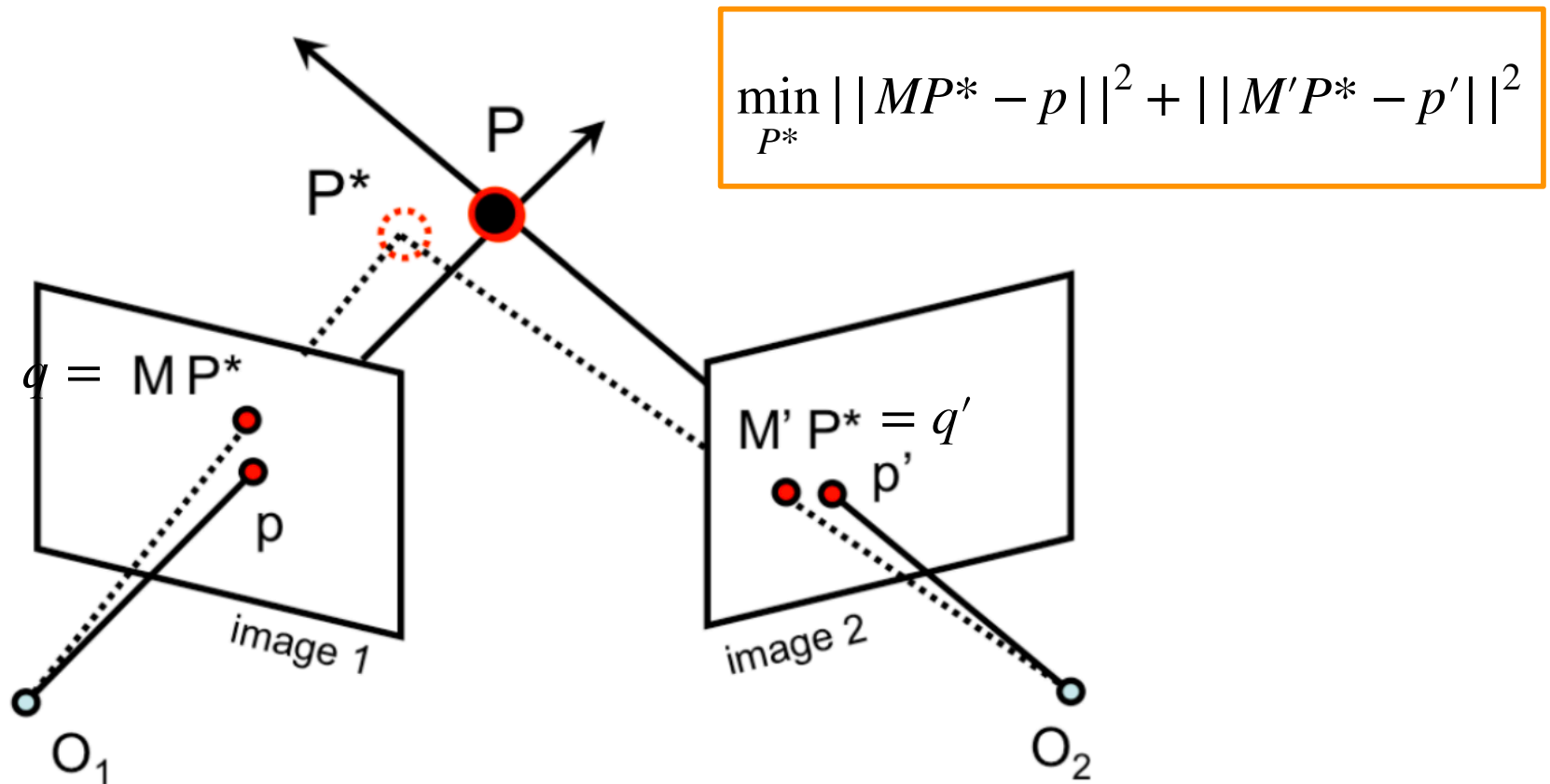
Triangulation: geometric approach

- ❖ **P** – mean point of the segment **AB** joining r_1, r_2 in the closest point
- ❖ **AB** perpendicular to both r_1, r_2



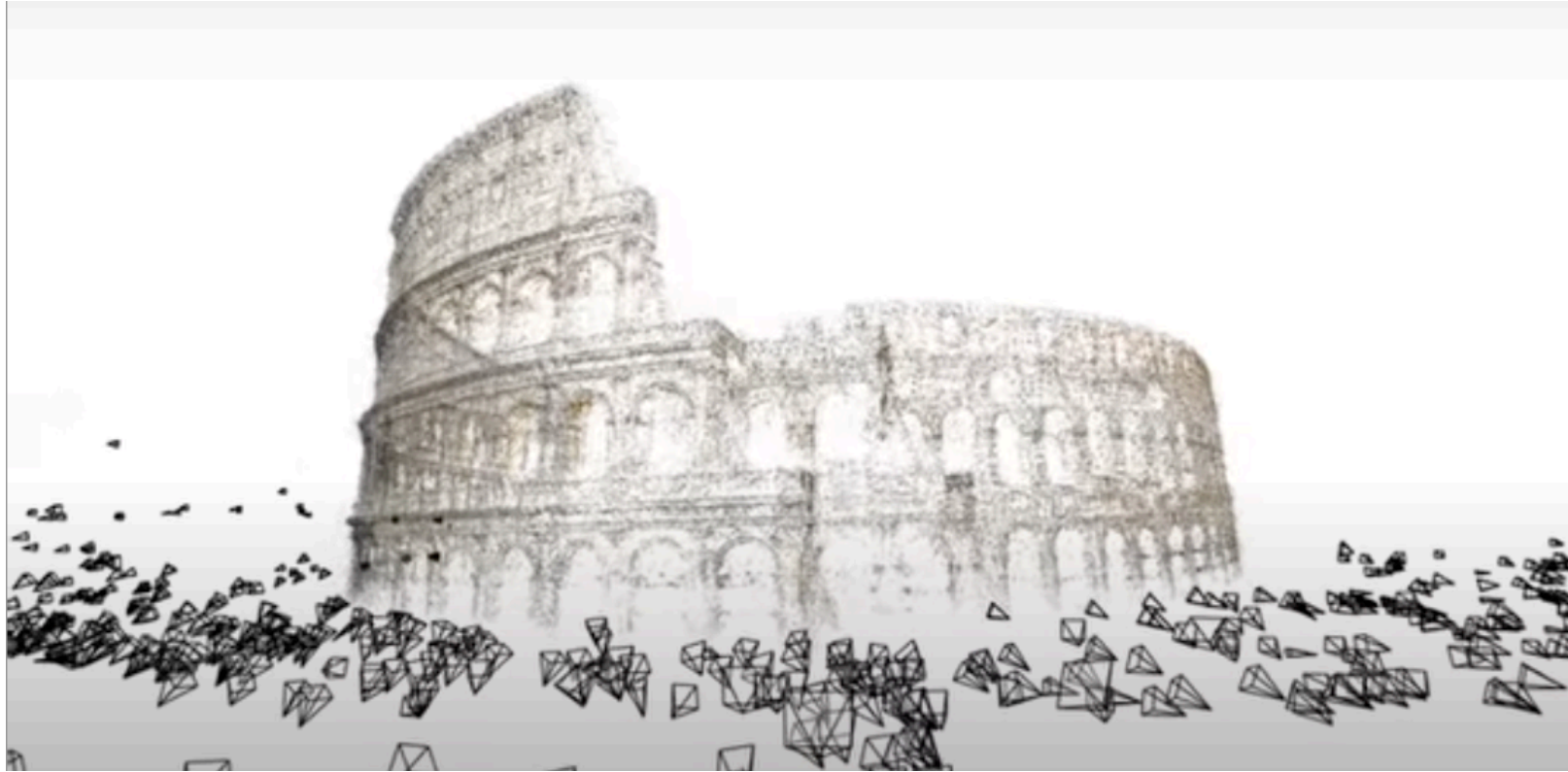
Triangulation: non linear algebraic approach

- ❖ Linear solution is often not the best
- ❖ $\rightarrow P^*$ – 3D point which projections q, q' , have minima quadratic distance from p, p' (different solution. eg: Gauss-Newton algorithm)



3D reconstruction with internet images

- ❖ <https://grail.cs.washington.edu/rome/index.html>



Lab time

- StereoVision